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TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	5
BACKGROUND	13
OBJECTIVES	14
METHODS	14
RESULTS	17
Figure 1.	18
Figure 2.	23
Figure 3.	25
DISCUSSION	29
AUTHORS' CONCLUSIONS	31
SUPPLEMENTARY MATERIALS	31
ADDITIONAL INFORMATION	32
REFERENCES	34
ADDITIONAL TABLES	38

[Intervention Review]

Minimally invasive trabecular meshwork surgery for open-angle glaucoma

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ABSTRACT

Rationale

Glaucoma is the leading cause of irreversible blindness. Minimally invasive glaucoma surgery (MIGS) techniques, including those that open or enlarge Schlemm's canal (SC) behind the trabecular meshwork (SC-MIGS), have been introduced to help optimise control of intraocular pressure (IOP), which may slow the progression of glaucoma. This review supersedes and expands on the scope of a previous Cochrane review on ab interno trabecular bypass surgery with Trabectome for open-angle glaucoma.

Objectives

To assess the effects of minimally invasive Schlemm's canal surgery in comparison to medical, laser, or surgical treatment (including other MIGS techniques) in people with open-angle glaucoma.

Search methods

We used CENTRAL, MEDLINE, Embase, two other databases, and trial registries, together with reference checking, citation searching, and contact with study authors, to identify studies that are included in the review. The latest search date was 12 October 2025.

Eligibility criteria

We searched for randomised controlled trials (RCTs) comparing SC-MIGS with other treatments (other MIGS, laser, phacoemulsification, or medical therapy). Reported SC-MIGS procedures included gonioscopy-assisted transluminal trabeculotomy, ab interno canaloplasty, Kahook Dual Blade goniotomy, excimer laser trabeculotomy, and ab interno trabeculectomy. We excluded studies of non-SC-MIGS

Minimally invasive trabecular meshwork surgery for open-angle glaucoma (Review)

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(e.g. iStent inject, Hydrus, or other microstents), conventional non-MIGS glaucoma surgery (e.g. trabeculectomy), and non-MIGS laser procedures (e.g. selective laser trabeculoplasty (SLT)). We included studies where procedures were combined with phacoemulsification and compared to phacoemulsification in combination with other glaucoma surgery or phacoemulsification alone.

Outcomes

Critical outcomes were the proportion of participants medication-free at each time point and the rate of visual field progression (decibels/time) or proportion of participants whose visual field loss progressed during follow-up. Important outcomes included mean change in IOP; mean change in IOP-lowering drops taken per day; proportion of participants requiring further glaucoma surgery or laser; and mean change in health-related quality of life. Adverse event outcomes included loss of visual acuity, bleeding, endophthalmitis, and IOP spikes. All outcomes were measured in the short term (6 to 18 months), medium term (> 18 to 36 months), and long term (> 36 months or longer).

Risk of bias

We assessed risk of bias using the Cochrane RoB 2 tool.

Synthesis methods

We used standard Cochrane methodology. We undertook meta-analysis where the data appeared to be clinically, methodologically, and statistically homogeneous. We used a random-effects model when there were three or more studies, otherwise we used a fixed-effect model. Where we could not perform a meta-analysis due to high heterogeneity, we provided a narrative summary of the data following guidance in the *Cochrane Handbook*.

Included studies

We included 10 parallel-group RCTs (639 participants). The mean age ranged from 41 to 74 years, and the proportion of male participants from 25% to 74%. Ethnicity was reported in five studies (Japanese (two), Caucasian/white (one), various ethnicities (two)).

One study was multicentre and one multinational; the remainder were single-centre studies. Studies were conducted in Italy (two), Japan (two), and India, the United States and Latin America, Poland, Canada, and China (one each).

Synthesis of results

There were five comparisons in this review.

For comparison 1 (SC-MIGS versus non-MIGS surgical options), there was no evidence of a short-term difference in medication-free participants (without phacoemulsification: risk ratio (RR) 1.15, 95% confidence interval (CI) 0.88 to 1.51; 1 study; low-certainty evidence; with phacoemulsification: RR 1.04, 95% CI 0.86 to 1.27; $I^2 = 39\%$; 2 studies; very low-certainty evidence), IOP change (without phacoemulsification: mean difference (MD) -0.75 mmHg, 95% CI -2.85 to 1.35; 1 study; low-certainty evidence; with phacoemulsification: MD 3.70 mmHg, 95% CI -3.25 to 10.65; 1 study; very low-certainty evidence), change in IOP-lowering drops per day (with phacoemulsification: MD -0.71, 95% CI -1.79 to 0.37; 1 study; very low-certainty evidence), further surgery (with phacoemulsification: RR 2.70, 95% CI 0.13 to 58.24; 2 studies; very low-certainty evidence), or adverse events. One study narratively reported that there were no changes in the rate of visual field progression.

For comparison 2 (SC-MIGS with phacoemulsification versus phacoemulsification only), in the short term, SC-MIGS may increase medication-free participants (RR 105.00, 95% CI 6.64 to 1660.79; 1 study; low-certainty evidence) and result in a greater mean IOP change (MD -8.70 mmHg, 95% CI -10.12 to -7.28; 1 study; low-certainty evidence). There was no evidence of a difference in further surgery (RR 0.33, 95% CI 0.02 to 7.24; 1 study; very low-certainty evidence). SC-MIGS may increase bleeding risk (RR 13.42, 95% CI 1.86 to 96.98; 2 studies; low-certainty evidence), with no evidence of a difference in other adverse events.

For comparison 3 (SC-MIGS versus non-SC-MIGS), there was no evidence of a difference in medication-free participants (RR 1.07, 95% CI 0.92 to 1.24; 2 studies; very low-certainty evidence), IOP change (MD -0.25, 95% CI -1.13 to 0.63; 2 studies; very low-certainty evidence), change in IOP-lowering drops per day (MD -0.15, 95% CI -0.34 to 0.03; 2 studies; very low-certainty evidence), further surgery (no participants in either group), or adverse events.

Comparison 4 (SC-MIGS versus other SC-MIGS) showed there were likely no differences between the groups in nearly all outcomes.

Comparison 5 (SC-MIGS versus laser) showed there may be a greater reduction in mean IOP, but there may be a greater risk of bleeding in the SC-MIGS group.

Authors' conclusions

The available evidence is of low or very low certainty, and we are therefore able to draw only weak, uncertain conclusions about the effects of SC-MIGS. There may be little to no difference in the proportion of participants who are medication-free in the short, medium, or long term between SC-MIGS and non-MIGS surgical treatment. SC-MIGS may demonstrate a higher proportion of participants to be medication-free in the short term compared to phacoemulsification surgery alone; however, there may be a higher risk of bleeding. SC-MIGS may also be associated with a greater reduction in mean IOP compared to laser treatment.

Minimally invasive trabecular meshwork surgery for open-angle glaucoma (Review)

Funding

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Registration

Protocol (2022): doi.org/10.1002/14651858.CD015186

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of minimally invasive Schlemm's canal surgery for treating the common eye condition glaucoma?

Key messages

- When types of minimally invasive glaucoma surgery that act on Schlemm's canal (called SC-MIGS) are compared with non-minimally invasive (conventional) glaucoma surgeries, there may be little to no difference in the proportion of people who are medication-free in the short, medium, or long term.
- More people may be medication-free in the short term after SC-MIGS compared with cataract surgery alone; however, there may be a higher risk of bleeding.
- SC-MIGS may reduce eye pressure more than laser treatment.

What is open-angle glaucoma?

Open-angle glaucoma is a common eye condition in which there is damage to the optic nerve, which usually progresses slowly over many years. 'Open-angle' means that the eye's natural drainage channels are not blocked but do not drain fluid effectively. This can increase pressure in the eye and damage the optic nerve, causing vision loss, often affecting the peripheral vision before the central vision. Some people develop glaucoma despite normal eye pressure and require further pressure-lowering treatment.

How is open-angle glaucoma treated?

Treatment includes eye drops, laser treatment, and surgery. Surgery is usually used if eye drops or laser treatment has not worked.

Standard surgery usually involves bypassing the drainage channels of the eye to improve fluid drainage. It may be performed under local anaesthetic (while the patient is awake) or general anaesthetic (while the patient is asleep). Complications, such as bleeding in the eye or very low eye pressure, can occur, and some people may need repeat surgery.

One alternative to standard surgical procedures is minimally invasive glaucoma surgery. Some of these procedures act on a part of the drainage network of the eye called Schlemm's canal.

What did we want to find out?

We wanted to find out whether these types of minimally invasive glaucoma surgery that specifically act on Schlemm's canal (called SC-MIGS below) help improve outcomes and whether they cause any harmful or unwanted effects in people with open-angle glaucoma. We were particularly interested in:

- the likelihood of people not needing eye drops;
- the likelihood of progressive visual field loss (loss of central or side areas of vision);
- changes in eye pressure;
- changes in the number of eye drops used following surgery;
- any unwanted effects or side effects.

What did we do?

We searched for studies that compared SC-MIGS to other types or combinations of surgery, or to laser treatment. We compared and summarised the results of the studies and rated our confidence in the evidence, based on factors such as study methods and sizes.

What did we find?

We found 10 studies that involved 639 people (639 eyes). People in the studies were men and women aged between 41 and 74 years. The studies took place in Italy, Japan, India, the USA and Latin America, Poland, Canada, and China. The studies compared:

- SC-MIGS with conventional glaucoma surgery;
- SC-MIGS plus cataract surgery with cataract surgery alone;
- SC-MIGS with types of minimally invasive glaucoma surgery that don't act on Schlemm's canal;

Minimally invasive trabecular meshwork surgery for open-angle glaucoma (Review)

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- one type of SC-MIGS with another type of SC-MIGS;
- SC-MIGS with laser procedures.

SC-MIGS versus standard glaucoma surgery (3 studies, 99 people)

There may be no difference between SC-MIGS and standard glaucoma surgery in the likelihood of people not needing eye drops to control their glaucoma, having progressive visual field loss, or in the average change in eye pressure, number of eye drops people used following surgery, or unwanted effects.

SC-MIGS plus cataract surgery versus cataract surgery alone (2 studies, 132 people)

People may be less likely to need eye drops to control their glaucoma, average change in eye pressure may be greater, but there may be more risk of bleeding with SC-MIGS plus cataract surgery compared with cataract surgery alone.

SC-MIGS versus types of minimally invasive glaucoma surgery that don't act on Schlemm's canal (2 studies, 238 people)

There may be no difference between these two treatments in how likely people are not to need eye drops to control their glaucoma, number of eye drops used following surgery, or need for further surgery.

One type of SC-MIGS versus another type of SC-MIGS (2 studies, 140 people)

There is likely no difference between one type of SC-MIGS and another in any of the things we wanted to find out.

SC-MIGS versus laser procedures (1 study, 30 people)

Eye pressure may be reduced more, but there may be more risk of bleeding with SC-MIGS compared with laser procedures.

What are the limitations of the evidence?

Our confidence in the evidence is very limited because there were not many studies, and they did not include many people. The studies did not use the best methods, so their results may not be reliable. Also, the studies each focused on a small group of people of the same ethnicity or nationality, so we don't know how their findings will apply to different groups of people.

How up to date is this evidence?

The evidence is up to date as of October 2025.

SUMMARY OF FINDINGS

Summary of findings 1. SC-MIGS (MIGS that act on the Schlemm's canal) compared with non-MIGS surgical options for open-angle glaucoma

SC-MIGS (MIGS that act on the Schlemm's canal) compared with non-MIGS surgical options for open-angle glaucoma						
Patient or population: people with open-angle glaucoma						
Settings: hospitals or eye clinic						
Intervention: SC-MIGS						
Comparison: non-MIGS surgical options						
Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Non-MIGS	SC-MIGS				
Proportion of participants who were medication-free at short-term follow-up (6 to 18 months)	Single procedure					
	813 per 1000	934 per 1000 (715 to 1000)	RR 1.15 (0.88 to 1.51)	32 (1 study)	⊕⊕⊕⊖ Low ^a	—
	With phaco					
	875 per 1000	910 per 1000 (753 to 1000)	RR 1.04 (0.86 to 1.27)	66 (2 studies)	⊕⊕⊕⊖ Very low ^{a,b}	—
Rate of visual field progression (dB/time) at short-term follow-up (6 to 18 months)	—				⊕⊕⊕⊖ Very Low ^{c,d}	Kicinska 2023 narratively reported that there were no differences between the groups when comparing the MD and the PSD parameters.
Mean change in IOP at short-term follow-up (6 to 18 months)	Single procedure					—

Lower IOP is favoured (i.e. greater negative mean difference is favoured)	Mean reduction in IOP in the SC-MIGS group was on average 0.75 mmHg (2.85 to -1.35) greater than in the non-MIGS group.		MD -0.75 (-2.85 to 1.35)	32 (1 study)	⊕⊕⊕⊖ Low^a	
	With phaco					
	Mean reduction in IOP in the SC-MIGS group was on average 3.70 mmHg (3.25 to -10.65) lower than in the non-MIGS group.		MD 3.70 (-3.25 to 10.65)	17 (1 study)	⊕⊕⊕⊖ Very low^{a,b}	
Mean change in number of IOP-lowering drops taken per day at short-term follow-up (6 to 18 months) Lower number of IOP-lowering drops taken per day is favoured (i.e. greater negative mean difference is favoured)	With phaco					
	Mean reduction in number of IOP-lowering drops taken per day in the SC-MIGS group was on average 0.71 (1.79 to -0.37) lower than in the non-MIGS group.		MD -0.71 (-1.79 to 0.37)	16 (1 study)	⊕⊕⊕⊖ Very low^{a,b}	—
Proportion of participants who required further glaucoma surgery (including laser) at short-term follow-up (6 to 18 months)	Single procedure					
	0 per 1000	0 per 1000	Not estimable	32 (1 study)	⊕⊕⊕⊖ Low^a	—
	With phaco					
	0 per 1000	40 per 1000 (5 to 1000)	RR 2.70 (0.13 to 58.24)	65 (2 studies)	⊕⊕⊕⊖ Very low^{a,b}	—
Mean change in HRQoL at short-term follow-up (6 to 18 months)	No studies reported this outcome.					
Proportion of participants experiencing intra- and postoperative complications (any time point)	Loss of visual acuity					
	No studies reported this outcome.					
	Bleeding – single procedure					
	0 per 1000	375 per 1000 (296 to 1000)	RR 13.00 (0.79 to 213.09)	32 (1 study)	⊕⊕⊕⊖ Low^a	—
	Bleeding – with phaco					

550 per 1000	622 per 1000 (429 to 891)	RR 1.13 (0.78 to 1.62)	65 (2 studies)	⊕⊕⊕⊕ Very low^{a,b}	—
IOP spikes – with phaco					
150 per 1000	216 per 1000 (87 to 531)	RR 1.44 (0.58 to 3.54)	65 (2 studies)	⊕⊕⊕⊕ Very low^{a,b}	—
Endophthalmitis – single procedure					
0 per 1000	0 per 1000	Not estimable	32 (1 study)	⊕⊕⊕⊕ Moderate^d	—
Endophthalmitis – with phaco					
0 per 1000	0 per 1000	Not estimable	48 (1 study)	⊕⊕⊕⊕ Moderate^d	—

*The basis for the **assumed risk** is the mean baseline risk from the studies in the meta-analysis; the total number of events in the control group divided by the total number of participants in the control groups, scaled to 1000. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HRQoL:** health-related quality of life; **IOP:** intraocular pressure; **MD:** mean difference; **MIGS:** minimally invasive glaucoma surgery; **Phaco:** phacoemulsification; **PSD:** Power Spectral Density; **RR:** risk ratio; **SC-MIGS:** MIGS that act on the Schlemm's canal

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded two levels for imprecision due to very few events and wide CI (–2).

^bDowngraded one level for risk of bias (–1).

^cDowngraded two levels for risk of bias (–2).

^dDowngraded one level for imprecision due to small sample size/small number of events/single study.

Summary of findings 2. SC-MIGS (MIGS that act on the Schlemm's canal) + phacoemulsification compared with phacoemulsification only for open-angle glaucoma

SC-MIGS (MIGS that act on the Schlemm's canal) + phacoemulsification compared with phacoemulsification only for open-angle glaucoma

Patient or population: people with open-angle glaucoma

Settings: hospitals or eye clinic

Intervention: SC-MIGS + phaco

Comparison: phaco only

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Phaco only	MIGS				
Proportion of participants who were medication-free at short-term follow-up (6 to 18 months)	0 per 1000	912 per 1000	RR 105.00 (6.64 to 1660.79)	114 (1 study)	⊕⊕⊕⊖ Low^a	—
Rate of visual field progression (dB/time) at short-term follow-up (6 to 18 months)	No studies reported this outcome.					
Mean change in IOP at short-term follow-up (6 to 18 months) Lower IOP is favoured (i.e. greater negative mean difference is favoured)	Mean reduction in IOP in the SC-MIGS group was on average 8.70 mmHg (7.28 to 10.12) greater than in the phaco-only group.		MD -8.70 (-10.12 to -7.28)	114 (1 study)	⊕⊕⊕⊖ Low^{b,c,d}	—
Mean change in number of IOP-lowering drops taken per day at short-term follow-up (6 to 18 months) Lower number of IOP-lowering drops taken per day is favoured (i.e. greater negative mean difference is favoured)	Mean reduction in number of IOP-lowering drops taken per day in the SC-MIGS group was on average 0.10 (0.39 to -0.19) greater than in the phaco-only group.		MD -0.10 (-0.39 to 0.19)	114 (1 study)	⊕⊕⊕⊖ Low^a	—
Proportion of participants who required further glaucoma surgery (including laser) at short-term follow-up (6 to 18 months)	111 per 1000	37 per 1000 (2 to 804)	RR 0.33 (0.02 to 7.24)	18 (1 study)	⊕⊕⊕⊖ Very low^{a,e}	—



Mean change in HRQoL at short-term follow-up (6 to 18 months)	No studies reported this outcome.					
Proportion of participants experiencing intra- and postoperative complications (any time point)	Loss of visual acuity					
	0 per 1000	0 per 1000	Not estimable	18 (1 study)	⊕⊕⊕⊕	— Very low ^{b,e}
	Bleeding					
	0 per 1000	152 per 1000	RR 13.42 (1.86 to 96.98)	132 (2 studies)	⊕⊕⊕⊕	— Low ^{e,f,g,h}
	IOP spikes					
	333 per 1000	333 per 1000 (90 to 1000)	RR 1.00 (0.27 to 3.69)	18 (1 study)	⊕⊕⊕⊕	— Very low ^{a,e}
	Endophthalmitis					
	0 per 1000	0 per 1000	Not estimable	18 (1 study)	⊕⊕⊕⊕	— Very low ^{b,e}

*The basis for the **assumed risk** is the mean baseline risk from the studies in the meta-analysis; the total number of events in the control group divided by the total number of participants in the control groups, scaled to 1000. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HRQoL:** health-related quality of life; **IOP:** intraocular pressure; **MD:** mean difference; **MIGS:** minimally invasive glaucoma surgery; **Phaco:** phacoemulsification; **RR:** risk ratio; **SC-MIGS:** MIGS that act on the Schlemm's canal

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded two levels for imprecision due to very few events and wide CI (−2).

^bDowngraded one level for imprecision due to very few events (−1).

^cDowngraded two levels for inconsistency due to very high I² statistic (−2).

^dUpgraded one level due to large magnitude of effect.
^eDowngraded two levels for risk of bias (−2).
^fUpgraded two levels due to very large magnitude of effect.
^gDowngraded one level for imprecision due to very wide confidence interval.
^hDowngraded one level for differences in intervention.

Summary of findings 3. SC-MIGS (MIGS that act on the Schlemm's canal) compared with non-SC-MIGS (MIGS that do not act on the Schlemm's canal) for open-angle glaucoma

SC-MIGS (MIGS that act on the Schlemm's canal) compared with non-SC-MIGS (MIGS that do not act on the Schlemm's canal) for open-angle glaucoma						
Patient or population: people with open-angle glaucoma						
Settings: hospitals or eye clinic						
Intervention: SC-MIGS						
Comparison: non-SC-MIGS						
Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Non-SC-MIGS	SC-MIGS				
Proportion of participants who were medication-free at short-term follow-up (6 to 18 months)	730 per 1000	782 per 1000 (672 to 905)	RR 1.07 (0.92 to 1.24)	227 (2 studies)	⊕⊕⊕⊕ Very low ^{a,b}	—
Rate of visual field progression (dB/time) at short-term follow-up (6 to 18 months)	No studies reported this outcome.					
Mean change in IOP at short-term follow-up (6 to 18 months) Lower IOP is favoured (i.e. greater negative mean difference is favoured)	Mean reduction in IOP in the SC-MIGS group was on average 0.25 mmHg (1.13 to -0.63) greater than in the non-SC-MIGS group.		MD -0.25 (-1.13 to 0.63)	227 (2 studies)	⊕⊕⊕⊕ Very low ^{a,b}	—
Mean change in number of IOP-lowering drops taken per day at short-term follow-up (6 to 18 months)	Mean reduction in number of IOP-lowering drops taken per day in the SC-MIGS group was on average 0.15 (0.34 to -0.03) greater than in the non-SC-MIGS group.		MD -0.15 (-0.34 to 0.03)	227 (2 studies)	⊕⊕⊕⊕ Very low ^{a,b}	—

Lower number of IOP-lowering drops taken per day is favoured (i.e. greater negative mean difference is favoured)						
Proportion of participants who required further glaucoma surgery (including laser) at short-term follow-up (6 to 18 months)	0 per 1000	0 per 1000	Not estimable	70 (1 study)	⊕⊕⊕⊖	— Very low^b
Mean change in HRQoL at short-term follow-up (6 to 18 months)	No studies reported this outcome.					
Proportion of participants experiencing intra- and postoperative complications (any time point)	Loss of visual acuity					
	0 per 1000	0 per 1000	Not estimable	166 (1 study)	⊕⊕⊕⊖	— Moderate^c
	Bleeding					
	8 per 1000	26 per 1000 (4 to 160)	RR 3.10 (0.50 to 19.36)	238 (2 studies)	⊕⊕⊕⊖	— Low^d
	IOP spikes					
	240 per 1000	237 per 1000 (153 to 364)	RR 0.99 (0.64 to 1.52)	238 (2 studies)	⊕⊕⊕⊖	— Moderate^a
*The basis for the assumed risk is the mean baseline risk from the studies in the meta-analysis; the total number of events in the control group divided by the total number of participants in the control groups, scaled to 1000. The corresponding risk (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI). CI: confidence interval; HRQoL: health-related quality of life; IOP: intraocular pressure; MD: mean difference; MIGS: minimally invasive glaucoma surgery; Phaco: phacoemulsification; RR: risk ratio; SC-MIGS: MIGS that act on the Schlemm's canal						
GRADE Working Group grades of evidence High certainty: we are very confident that the true effect lies close to that of the estimate of the effect. Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different. Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect. Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.						

^aDowngraded one level for imprecision due to CI crossing the threshold (−1).

^bDowngraded two levels for high risk of bias (−2).

- ^cDowngraded one level for imprecision due to very few events (–1).
- ^dDowngraded two levels for imprecision due to very few events and wide CI (–2).

BACKGROUND

Description of the condition

Glaucoma is a chronic progressive optic neuropathy affecting 3.5% of people between the ages of 40 and 80. It is the leading cause of irreversible blindness, affecting 76 million people globally in 2014; this is expected to increase to 111.8 million people by 2040 [1].

Open-angle glaucoma (OAG) is the most common type, with an estimated global prevalence of 2.4% that equates to an affected population of nearly 69 million people [2]. In one large population cohort, one in six people with OAG became bilaterally blind [3]. The only proven way to prevent vision loss is to reduce the pressure inside the eye (intraocular pressure (IOP)) over the long term [4, 5, 6, 7]. OAG may be classified as primary when there is no identifiable cause, or secondary when there is a known cause. Approaches to reducing IOP include medical therapy, laser treatments, and surgery.

As commercially available eye drop preparations have a short-lasting effect, medical therapy requires eye drops to be instilled one or more times daily for life. Adherence is very poor, even if use is monitored [8, 9]. Conventional surgical techniques such as trabeculectomy are associated with significant risks, with more than 40% of patients developing perioperative complications [10, 11]. Between 7% and 18% of cases have been shown to require reoperation [10, 12], with one recent single centre study reporting a rate of 23% [13]. These techniques are, therefore, often reserved for disease that is progressing despite other treatments [14]. Other treatments, such as selective laser trabeculoplasty [15] and new minimally invasive surgical techniques, are alternatives to conventional surgery or eye drop preparations.

Description of the intervention and how it might work

A number of minimally invasive surgical techniques have been developed with the aim of achieving long-term reduction of IOP by acting on the drainage apparatus of the eye (including trabecular meshwork and Schlemm's canal), with a better safety profile than conventional surgery [16]. Since the inception of our previous Cochrane review on ab interno trabecular bypass surgery with Trabectome [17], a new range of minimally invasive surgical techniques has emerged to lower IOP by acting on Schlemm's canal, either directly or indirectly. In this review, we expand our scope to include all minimally invasive procedures that open or enlarge Schlemm's canal without implanted hardware in open-angle glaucoma (SC-MIGS). As other Cochrane systematic reviews have already covered implanted MIGS devices that bypass the trabecular meshwork (e.g. iStent Inject) [18, 19, 20, 21], these were excluded from this review.

Some of the procedures that we anticipated being included were:

- Ab interno trabeculectomy with Trabectome: Trabectome uses a bipolar 550 kHz electrode to ablate the trabecular meshwork over 180°. The device utilises ionisation and disintegration, allowing any heat dissipation to be well confined [22].
- Gonioscopy-assisted transluminal trabeculotomy (GATT): the GATT procedure is a modification of the traditional trabeculotomy that involves cannulating and then unroofing the entire Schlemm's canal [23].
- Kahook Dual Blade goniotomy: this device for goniotomy is designed to allow for near-complete removal of the trabecular meshwork, compared with other techniques where residual trabecular meshwork may remain. Minimising these residues may result in less fibrosis and therefore better long-term outcomes compared with other procedures [24].
- Ab interno canaloplasty with iTrack: this combines catheterisation and pressurised viscodilation of Schlemm's canal, which separates compressed trabecular planes within the trabecular meshwork [25].
- Other variations of ab interno trabecular bypass surgery: for example, microhook ab interno trabeculotomy [26], needle goniotomy, or trabecular meshwork incision.

Some of these interventions are frequently combined with phacoemulsification (cataract surgery). Phacoemulsification in itself is known to reduce IOP [27]. Therefore, we planned to specifically examine the evidence for each of the treatments in people who had had concomitant phacoemulsification compared with those who had not.

One method to decrease the IOP would be to increase the outflow of aqueous humour (the clear fluid between the lens and cornea). Aqueous humour drains through several tissues, including the trabecular meshwork (which is where the largest resistance to flow occurs) and Schlemm's canal [28].

As such, methods to increase the outflow can include bypassing or removing the trabecular meshwork using ab interno Schlemm's canal surgery. Dilating Schlemm's canal causes compressed trabecular planes within the trabecular meshwork to be separated, which may contribute to increased aqueous outflow. The procedures introduced above are thought to work as follows:

- Trabectome: Trabectome is designed to selectively ablate a portion of the trabecular meshwork, enabling aqueous humour to have direct access to Schlemm's canal and thence the collector channels [29]. This is intended to promote aqueous outflow, thereby reducing IOP.
- GATT: after an incision through the trabecular meshwork via the cornea, Schlemm's canal is cannulated 360°. Schlemm's canal is unroofed, allowing the trabecular meshwork to be entirely bypassed [23].
- Kahook Dual Blade: the Kahook Dual Blade is designed specifically for a more complete removal of trabecular meshwork tissue than other goniotomy techniques. It has a sharp tip to allow for entry of the blade through the trabecular meshwork to Schlemm's canal. The device has a ramp, allowing for gentle stretching of the trabecular meshwork. As the blade advances, parallel incisions are made into the trabecular meshwork. This results in a strip of the trabecular meshwork being excised [24].
- OMNI surgical system: this allows for Schlemm's canal to be microcatheterised circumferentially from a single corneal incision. This can then be followed by a trabeculotomy using the same device [30].
- iTrack: by catheterisation and pressurised viscodilation of Schlemm's canal, compressed trabecular planes within the trabecular meshwork are separated and any herniated inner wall tissue is caused to withdraw from the collector channels [25].

Why it is important to do this review

Consultation with glaucoma patients and healthcare professionals has identified a need for better treatments for glaucoma [31, 32]. Minimally invasive glaucoma procedures may safely and effectively reduce IOP in the long term, offering another surgical option for people with open-angle glaucoma where medical therapy is ineffective, not tolerated, or not adhered to. A single treatment may also be more acceptable to people than daily and indefinite self-administration of eye drops [33, 34].

Results of several of these different Schlemm's canal surgeries have been reported. Positive initial results of ab interno trabecular bypass surgery with Trabectome were reported in 2005 [35]. There were 4600 treatments performed between 2004 and 2013 at over 200 centres worldwide [36, 37]. Results from studies of the Kahook Dual Blade [38, 39], the GATT procedure [40, 41], the OMNI surgical system [30], and ab interno canaloplasty with or without goniotomy using iTrack also show efficacy in reducing IOP [25], with positive surgical outcomes.

Therefore, due to the potential benefits of each of these procedures for people with open-angle glaucoma, it is important to evaluate the evidence to determine whether treatment with each procedure is safe and efficacious.

This Cochrane review supersedes and expands upon the published review on ab interno trabecular bypass surgery with Trabectome for open-angle glaucoma [17]. It is part of a suite of reviews on minimally invasive glaucoma surgery (MIGS) techniques and devices currently being undertaken by the Cochrane Eyes and Vision MIGS Consortium.

OBJECTIVES

To assess the effects of minimally invasive Schlemm's canal surgery in comparison to medical, laser, or surgical treatment (including other MIGS techniques) in people with open-angle glaucoma.

METHODS

This review relies on the same methodology as the original review on this topic but with a larger scope, which includes all MIGS surgeries that enlarge Schlemm's canal or open the trabecular meshwork without placement of a bypass stent. This is because MIGS that act by bypassing the trabecular meshwork (e.g. iStent Inject) have already been covered by previous Cochrane reviews. We originally referred to the interventions in this review as MIGS that act on the trabecular meshwork in our review protocol. However, to avoid confusion, particularly in comparisons where canal-based procedures (e.g. canaloplasty and goniotomy) are compared directly with TM-bypass stents (e.g. iStent, Hydrus), we have instead adopted the term "Schlemm's canal MIGS" (SC-MIGS) to describe minimally invasive procedures that enhance aqueous outflow via Schlemm's canal without implantation of a bypass stent or permanent hardware.

- The secondary outcomes, adverse effects, and all follow-up times have been harmonised with the other MIGS reviews that form part of this suite of reviews.
- The original published review included combination therapy with phacoemulsification as an intervention and for subgroup

analysis. After discussion within the review team and MIGS Consortium, we opted to include it as a separate comparison.

- We changed the **Objectives** to include anyone with open-angle glaucoma instead of restricting it to just those "whose condition is inadequately controlled with drops".

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) only. We included all study reports published in English (or which have been translated into English), irrespective of their publication status.

We did not include quasi-RCTs (in which allocation is decided by an approximation of randomisation such as allocation by patient ID number). We included studies that included more than one eye but excluded within-person studies because of the potential for comparator treatments (such as topical beta blockers) or surgical procedures used for one eye to influence the outcome in the other eye [42, 43].

Types of participants

We included RCTs where participants had open-angle glaucoma of any type, including primary and secondary open-angle glaucoma. We excluded closed-angle glaucoma. The difference between open-angle and closed-angle glaucoma is that, while both are diseases of increased IOP, closed-angle glaucoma is often due to an obstruction to the outflow of aqueous as a result of a narrow or closed angle between the iris and the cornea. Open-angle glaucoma, on the other hand, is due to an increase in the resistance to flow in the drainage channels of the trabecular meshwork without a narrow angle between the iris and the cornea [44].

We also included participants with ocular hypertension or possible glaucoma (suspects for glaucoma). We did not apply restrictions regarding location, setting, or demographic factors. For studies that included only a subset of relevant participants that met the inclusion criteria, we included only eligible participants. When necessary, we contacted the individuals or organisations responsible for studies to collect specific information. If this was unsuccessful, we excluded these studies.

Types of interventions

We compared minimally invasive glaucoma surgeries that acted on the Schlemm's canal (including ab interno canaloplasty, Kahook Dual Blade goniotomy, GATT, excimer laser trabeculotomy) with the following:

- non-MIGS glaucoma surgery (trabeculectomy);
- phacoemulsification only;
- MIGS techniques that work on pathways other than the Schlemm's canal;
- another type of Schlemm's canal MIGS technique, such as those listed above;
- laser treatment (selective laser trabeculoplasty or argon laser trabeculoplasty);

We also included trials in which these techniques were combined with phacoemulsification compared with phacoemulsification in combination with other glaucoma surgery or alone. While we initially also planned to include studies where SC-MIGS was

compared with medical therapy alone, there were no studies found on this comparison during the screening process.

Outcome measures

Ideally, studies would have reported outcomes as per the recommendations of the World Glaucoma Association (WGA) [45]; however, we did not use the reporting of particular outcomes as a criterion for eligibility for this review. We did not exclude studies from this review solely on the grounds of an outcome of interest not being reported.

We reported outcomes in the short term (6 to 18 months), medium term (18 to 36 months), and long term (36 months or longer).

Critical outcomes

- Proportion of participants who are medication-free (not using eye drops) at each time point (short term, medium term, and long term).

Several glaucoma outcome measures have been specified as primary outcomes in other Cochrane reviews and protocols [46]. A recent study classified IOP, visual field, safety, and anatomic outcomes as being highly important to glaucoma experts [47]. A panel of patients from the Patient and Public Involvement Group of the National Institute for Health Research (NIHR) Biomedical Research Centre for Ophthalmology identified drop-free disease control as a highly valued outcome [48]. We chose a patient-centred critical outcome, in addition to one identified by glaucoma experts. In assessing this effect measure, we reported how prescription of IOP-lowering eye drops was determined during follow-up. We examined whether the people measuring IOP and those deciding on prescription of IOP-lowering eye drops were masked to the treatment group.

- Rate of visual field progression (decibels (dB)/time) or proportion of participants whose field loss progressed in the follow-up period.

Important outcomes

- Mean change in IOP, measured using Goldmann applanation tonometry. Goldmann applanation tonometry is the gold standard method of measuring IOP worldwide, and therefore this was the only appropriate measurement method that we accepted.
- Mean change in number of IOP-lowering drops taken per day.
- Proportion of participants who achieve an IOP of 21 mmHg or less.
- Proportion of participants who achieve an IOP of 17 mmHg or less.
- Proportion of participants who achieve an IOP of 14 mmHg or less.
- Proportion of participants who required further glaucoma surgery, including laser, as recorded by the investigators of the included trial.
- Mean change in health-related quality of life (HRQoL), measured using any scale.

Adverse effects

Proportion of participants experiencing intra- and postoperative complications, including, but not restricted to, the following:

- bleeding, as recorded by the investigators;
- IOP spikes (postoperative rise in IOP, measured using Goldmann applanation tonometry, of more than 10 mmHg compared with the previous assessment, including measurements taken during the first postoperative month);
- endophthalmitis, as recorded by the investigators;
- loss of visual acuity (more than two Snellen lines or more than 0.3 logMAR, according to the method of recording visual acuity, or loss of light perception).

Search methods for identification of studies

Electronic searches

The Cochrane Eyes and Vision Information Specialist conducted systematic searches in the following databases for randomised controlled trials and controlled clinical trials on 12 October 2025. There were no restrictions to language or year of publication for the searches. However, only studies with reports published in English (or which have been translated into English) were included.

- Cochrane Central Register of Controlled Trials (which contains the Cochrane Eyes and Vision Trials Register) (2025; Issue 10), in the Cochrane Library ([Supplementary material 1](#)).
- MEDLINE Ovid (1946 to October 2025) ([Supplementary material 1](#)).
- Embase Ovid (1980 to October 2025) ([Supplementary material 1](#)).
- ISRCTN registry (www.isrctn.com/editAdvancedSearch) ([Supplementary material 1](#)).
- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (www.clinicaltrials.gov) ([Supplementary material 1](#)).
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) (www.who.int/ictpr) ([Supplementary material 1](#)).

Searching other resources

We searched the reference lists of included studies for other possible studies, and contacted any individuals or organisations we believed may have conducted or be conducting relevant RCTs. We also searched the websites of the various manufacturers for any information on forthcoming trials. This includes:

- OMNI Surgical Device - Sight Sciences Inc, Menlo Park, CA, USA (www.sightsciences.com; www.omnisurgical.com);
- Kahook Dual Blade - New World Medical, Rancho Cucamonga, CA (<https://www.newworldmedical.com/product/kdb-glide/>);
- iTrack - Nova Eye Medical, Kent Town, SA, Australia (<https://www.nova-eye.com/e-unsoverview/itrack/>).

Prior to publication, we searched MEDLINE, Embase, and Retraction Watch (retractionwatch.com/), to ensure that none of the included studies have been withdrawn owing to errors or fraud.

Data collection and analysis

Selection of studies

We used Covidence to screen the search results [49]. Two review authors (KV, RG) independently screened titles and abstracts; if abstracts were not available, we screened full-text articles. Two

review authors (KV, RG) independently assessed full-text reports of all potentially eligible studies. In case of disagreement regarding eligibility, a third review author (KH) arbitrated. We placed any full-text reports that we rejected at this stage in the excluded studies table and recorded a reason for exclusion.

Data extraction and management

Two review authors (KV, RG) independently extracted study characteristics from reports of each study, as per [Supplementary material 9](#), and presented these in a 'Characteristics of included studies' table in RevMan [50]. Two authors (KV, RG) independently extracted data for the analyses. The data extraction form in [Supplementary material 9](#) was previously used in the published Cochrane review on ab interno trabecular bypass surgery with Trabectome for open-angle glaucoma [17], which this review supersedes.

If data from included studies were missing or unclear, we contacted the individuals or organisations involved to obtain clarification. We intended to collect and use the most detailed numerical data available to facilitate analyses of included studies. We obtained these data from individuals or organisations instead of using less precise methods, such as extracting numeric data from graphs. If this was necessary, two review authors (KV, RG) independently extracted the data.

In any cases where the two authors (KV, RG) were in disagreement with respect to any independent data extraction, a third author (KH) arbitrated.

Risk of bias assessment in included studies

Two review authors (KV, RG) independently assessed the risk of bias in each included trial using the Cochrane risk of bias tool (RoB 2) as described in Chapter 8 of the *Cochrane Handbook for Systematic Review of Interventions* [51]. We assessed the effect of assignment to intervention (intention-to-treat effect), regardless of adherence to the assigned intervention. We considered the following domains:

1. bias arising from the randomisation process;
2. bias introduced by deviations from intended interventions;
3. bias arising from missing outcome data;
4. bias in measurement of the outcome;
5. bias in selection of the reported result.

We applied the RoB 2 tool to the outcomes specified in the summary of findings tables. For these outcomes, the time points were defined as at short-term follow-up (6 to 18 months post-intervention), except for the proportion of participants experiencing intraoperative or postoperative complications, which was assessed at any time point.

Outcomes assessed were:

- proportion of participants who were medication-free;
- rate of visual field progression;
- mean change in IOP;
- mean change in the number of IOP-lowering drops taken per day;
- proportion of participants who required further glaucoma surgery;
- mean change in HRQoL (no studies reported this outcome);

- proportion of participants experiencing intra- and postoperative complications.

Two review authors (KV, RG) judged the overall risk of bias for each study as:

- low risk of bias: low risk of bias in all domains;
- some concerns: some concerns in at least one domain, but no domain is at high risk of bias;
- high risk of bias: high risk of bias in at least one domain, or some concerns in multiple domains (≥ 3).

If there were any disagreements between the two authors, a third author (KH) arbitrated.

Measures of treatment effect

We reported outcomes in the short term (6 to 18 months), medium term (>18 to 36 months), and long term (>36 months or longer) after randomisation.

There were two critical outcomes to this study. Firstly, for the proportion of participants who were medication-free, we used the risk ratio (RR) as the treatment effect measure. Secondly, for the rate of visual field progression, the mean difference (MD) was the treatment effect measure. Where studies reported the proportion of participants whose field loss progressed in the follow-up period, we used the odds ratio (OR) as the treatment effect measure.

We reported MD in IOP and MD in the number of IOP-lowering drops taken per day. We further reported the proportion of participants achieving various target IOPs or requiring further glaucoma surgery using ORs as the treatment effect measure. We also reported HRQoL outcomes as differences in means or ORs for continuous and binary data, respectively. We reported secondary safety outcomes as ORs.

Unit of analysis issues

We noted whether the studies included one or two eyes from each participant and whether randomisation had been conducted at the level of the participant or the eye. There is a potential for medical treatments, such as topical beta blockers, used for one eye to influence the outcome in the other eye [42]. Surgery to lower the IOP in one eye may also affect the IOP of the other eye [43]. We therefore excluded studies that adopted a paired-eye design. In one study (Falkenberg 2020 [52, 53]), while the methods allowed inclusion of two eyes per participant, the study authors confirmed that only one eye per participant was included in the analyses. Therefore, we included this study in the analysis of this review.

If we identified a multiple-arm study, we included it if the study design ensured that an independent analysis of each treatment group occurred.

Dealing with missing data

We aimed to minimise missing outcome data by contacting individuals and organisations to try to obtain them. If the data were unavailable, but the level of missing data in each group and reasons for missing data in each group were similar, we simply analysed available case data if an intention-to-treat (ITT) analysis was not performed. If the authors conducted their own ITT analysis despite missing data, we aimed to document whether they provided any justification for the method they used to deal with missing data,

and whether they compared their ITT result with an available case result.

Some outcomes that were mentioned in our protocol, such as 'Mean change in IOP' or 'Mean change in number of IOP-lowering drops taken per day', may not be reported by the authors of the included studies. Therefore, where possible, we calculated these outcomes. For the mean change, the value of IOP or the number of IOP-lowering drops taken per day at baseline was subtracted from the relevant follow-up values. We calculated the standard deviation for these data using the equation advised in the *Cochrane Handbook for Systematic Review of Interventions* with an imputed correlation coefficient of 0.5 as suggested by a senior author (GV) [54].

Reporting bias assessment

We aimed to use a funnel plot to assess the risk of publication bias if more than 10 trials were included in the analysis.

Synthesis methods

We performed quantitative syntheses when the included studies appeared to be clinically and statistically homogeneous. We checked that participants, interventions, comparators, and outcomes were sufficiently similar to give a clinically meaningful result, and that our I^2 value result indicated little inconsistency (that is, $I^2 < 50\%$). I^2 values of greater than 50% are indicative of substantial heterogeneity and are, therefore, suggestive that meta-analysis might not be wise. However, we would have considered the consistency of the effect estimates; if all estimates were in the same direction, we considered meta-analysis even where heterogeneity was evident, but commented on this. We used a random-effects model unless there were fewer than three eligible studies, in which case we used a fixed-effect model. Where we could not perform a meta-analysis due to heterogeneity, we performed a narrative summary of the data following the guidance in Chapter 12 of the *Cochrane Handbook for Systematic Review of Interventions* [55].

We aimed to assess heterogeneity between trials by careful examination of the study reports, assessing forest plots, and an examination of the I^2 value with its 95% confidence interval. We considered I^2 values greater than 50% as indicating substantial heterogeneity.

Investigation of heterogeneity and subgroup analysis

We did not plan to conduct subgroup analysis. However, when subgroups were used by investigators, we reported these subgroups.

Equity-related assessment

Equity was not assessed in this review.

Sensitivity analysis

When there were sufficient studies (> 10 for each outcome), we aimed to assess the impact of including studies at high risk of

bias by removing these from the sensitivity analysis and comparing the result with the primary analysis. However, we did not perform sensitivity analysis due to the small number of studies included for each review outcome.

Certainty of the evidence assessment

We prepared a table to summarise the findings of the review, including the assessment of the certainty of evidence for all outcomes using the GRADE approach outlined in Chapter 14 of the *Cochrane Handbook for Systematic Review of Interventions* [56]. For this, two authors (KV and RG) independently utilised the five GRADE considerations: risk of bias, consistency of effect, imprecision, indirectness, and publication bias.

We reported on the following outcomes at short-term follow-up (6 to 18 months) in a summary of findings table for each comparison listed in the [Types of interventions](#):

- proportion of participants who were medication-free (not using eye drops);
- rate of visual field progression (dB/time) or, secondarily, proportion of participants whose field loss progressed in the follow-up period;
- mean change in IOP, measured using Goldmann applanation tonometry;
- mean change in number of IOP-lowering drops taken per day;
- proportion of participants who required further glaucoma surgery, including laser;
- mean change in HRQoL;
- proportion of participants experiencing intra- and postoperative complications (any time point).

We planned to pool the various types of Schlemm's canal surgery in a single summary of findings table. However, in the case of strong evidence, ideally in the form of a randomised controlled trial, to suggest that different types of Schlemm's canal surgery yield different results, we planned to use separate tables.

Consumer involvement

Consumers were not involved in this review.

RESULTS

Description of studies

Results of the search

Our searches of the electronic databases on 12 October 2025 yielded a total of 1928 records. After duplicates had been removed, we screened 1696 titles and abstracts and retrieved 75 full-text reports (88 records) for further review. After excluding 39 studies (44 records, reasons reported in [Supplementary material 3](#)), we included 10 studies (18 records), identified 15 ongoing studies ([Supplementary material 5](#)), and listed 11 studies as awaiting classification ([Supplementary material 4](#)). A study selection flow diagram is shown in [Figure 1](#).

Figure 1. Study flow diagram

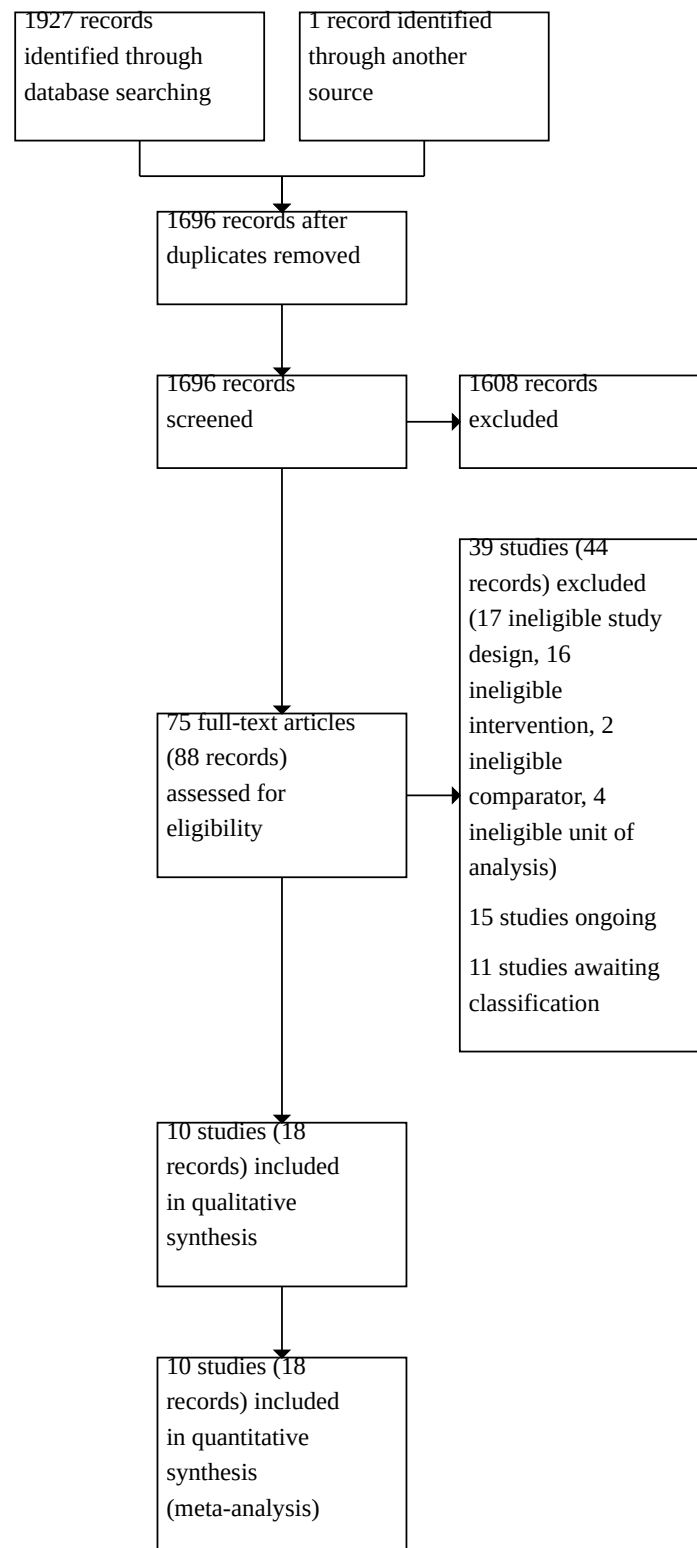


Figure 1. (Continued)

We are not aware of any other individuals or organisations who have conducted or may be conducting relevant RCTs.

We contacted one study investigator to confirm whether one or two eyes per participant were included in the analysis (Falkenberry 2020).

Included studies

A summary of all included studies is provided in Table 1. Further comprehensive information about each included study is available in the 'Characteristics of included studies' table (Supplementary material 2).

Types of studies

We included 10 completed parallel-group RCTs. A total of 639 eyes corresponding to 639 participants were initially randomised within these trials. At the end of the follow-up period for all the included studies, there were a total of 605 eyes analysed, corresponding to at least 605 participants.

Trial size varied from 18 eyes (Sato 2018 [57, 58]) to 166 eyes (Falkenberry 2020) at randomisation with a median of 54 eyes (54 participants) per study.

Two studies were multicentre (Falkenberry 2020; Goldberg 2024 [59]). All other included studies were single-centre.

There was one multinational study included (Goldberg 2024). Two studies each were conducted in Italy (Babighian 2010 [60]; Quaranta 1999 [61, 62]) and Japan (Arimura 2023 [63]; Sato 2018). One study each was conducted in India (Maheshwari 2023 [64]), China (Yin 2024 [65, 66]), the United States (Falkenberry 2020), Poland (Kicinska 2023 [67, 68, 69, 70]), and Canada (Ting 2018 [71, 72]).

Four studies did not declare a funding source (Babighian 2010; Maheshwari 2023; Quaranta 1999; Sato 2018). Three studies were funded by government grants (Arimura 2023; Kicinska 2023; Yin 2024). One study was funded by a university or hospital (Ting 2018). Finally, two studies were funded by industry (Falkenberry 2020; Goldberg 2024).

Types of participants

All studies included both male and female participants, though this was not specified by Quaranta 1999. The percentage of male participants ranged from 25% (Kicinska 2023) to 74% (Yin 2024). The mean age of participants in the included studies ranged from 41 years (Yin 2024) to 74 years (Sato 2018).

All studies specified that participants should be diagnosed with open-angle glaucoma (OAG) in their eligibility criteria. Five studies stated that they included participants with pseudoexfoliative glaucoma (Arimura 2023; Falkenberry 2020; Maheshwari 2023; Sato 2018; Ting 2018). Babighian 2010 and Yin 2024 included juvenile OAG. Maheshwari 2023 also specified ocular hypertension. Falkenberry 2020 also specified pigmentary glaucoma.

All studies required their participants to have had no previous glaucoma surgery.

All included studies enrolled adult participants. Four studies specified thresholds for IOP to further determine eligibility. Kicinska 2023 and Quaranta 1999 specified this as an IOP > 21 mmHg. Yin 2024 stated this to be < 22 mmHg if there was poor medication compliance or tolerance, otherwise > 22 mmHg and poorly controlled by medication. Falkenberry specified an IOP range of between 14 mmHg and 28 mmHg.

Three studies required participants to be on medication before being eligible – this was one medication or more for Quaranta 1999 and between one and three for Falkenberry 2020 and Goldberg 2024.

All included studies had one eye per participant that was eligible for the study. One study initially deemed one or two eyes per participant would be enrolled; however, after randomisation, only one eye per participant was enrolled (Falkenberry 2020).

Participant ethnic groups were specifically mentioned in five studies. These were Japanese (Arimura 2023; Sato 2018), Caucasian/white (Quaranta 1999), and several different ethnicities (Goldberg 2024; Ting 2018).

Types of interventions

We included studies that compared minimally invasive glaucoma surgeries that specifically act on the Schlemm's canal in open-angle glaucoma with other types of MIGs (including those that do not necessarily act on the Schlemm's canal), other types of glaucoma procedures that might not be minimally invasive, laser procedures, phacoemulsification, or medical therapy. We did not come across any studies where SC-MIGS were compared to medical therapy. We did not include studies where the primary intervention was a MIGS procedure that used a persistent implant to bypass the trabecular meshwork: such devices are covered by other Cochrane reviews [19, 20].

The interventions that have been included in this review following the systematic search are as follows:

- Excimer laser trabeculostomy (ELT) – a xenon chloride excimer laser is used to create channels in the trabecular meshwork into the Schlemm's canal, therefore increasing aqueous drainage.
- Canaloplasty ab interno with iTrack – a catheter is inserted into the Schlemm's canal and progressed through to up to 360 degrees ab interno. Following this, the surgeon withdraws the external segment of the catheter while injecting a viscoelastic through the tip to dilate the Schlemm's canal.
- Canaloplasty ab interno with STREAMLINE – a sleeve is depressed against the trabecular meshwork, following which a cannula enters the Schlemm's canal and injects viscoelastic into it. This is repeated three to eight times per eye at various points.
- Kahook Dual Blade (KDB) goniotomy – a goniotomy blade designed to allow for a near-complete removal of the trabecular

meshwork compared with other techniques, where residual trabecular meshwork may remain.

- Ab interno goniotrabeculotomy.
- Ab interno trabeculotomy with sutures.
- Ab interno trabeculotomy with Trabectome.
- Ab interno trabeculotomy with Tanito microhook.
- GATT – a modification of the traditional trabeculotomy that involves cannulating the entire Schlemm's canal, allowing access to the entire drainage system.

Types of outcomes

We included trials that fulfilled the inclusion criteria for study design, participants, interventions, and comparators, irrespective of whether our prespecified outcome data had been reported or not. Below is a list of our prespecified outcome data, and the studies that have reported these data.

Critical outcomes

- Proportion of participants who are medication-free (not using eye drops) at each time point (short term, medium term, long term).
 - Reported in the short term by all but two studies (Arimura 2023; Babighian 2010).
 - Reported in the medium term by five studies (Babighian 2010; Goldberg 2024; Kicinska 2023; Quaranta 1999; Sato 2018).
 - Reported in the long term by only one study (Kicinska 2023).
- Rate of visual field progression or proportion of participants whose field loss progressed in the follow-up period.
 - Reported by two studies (Babighian 2010; Kicinska 2023).

Important outcomes

- Mean change in IOP, measured using Goldmann applanation tonometry.
 - Reported by two studies only (Ting 2018; Yin 2024). It was calculated manually by ourselves in all other studies that included pre- and post-intervention IOP (Babighian 2010; Falkenberry 2020; Goldberg 2024; Maheshwari 2023; Quaranta 1999; Sato 2018).
- Mean change in the number of IOP-lowering drops taken per day.
 - Not reported by any study. It was manually calculated in other studies that included pre- and post-intervention number of IOP-lowering drops taken per day (Falkenberry 2020; Goldberg 2024; Maheshwari 2023; Ting 2018; Yin 2024).
- Proportion of participants who achieve an IOP of 21 mmHg or less.
 - Reported by two studies (Ting 2018; Yin 2024).
 - Falkenberry 2020 reported the proportion of participants who achieve an IOP of 22 mmHg or less.
- Proportion of participants who achieve an IOP of 17 mmHg or less.
 - Reported by only one study (Ting 2018).
 - Yin 2024 reported the proportion of participants who achieve an IOP of 15 mmHg or less and 18 mmHg or less.
- Proportion of participants who achieve an IOP of 14 mmHg or less.
 - Reported by three studies (Quaranta 1999; Sato 2018; Ting 2018).

- Proportion of participants who required further glaucoma surgery, including laser, as recorded by the investigators of the included trial.
 - Reported by four studies (Kicinska 2023; Quaranta 1999; Sato 2018; Ting 2018).
- Mean change in health-related quality of life (HRQoL), measured using any scale.
 - Not reported by any study.

Adverse events

Proportion of participants experiencing any intra- or postoperative complications, including the following:

- Loss of visual acuity (more than two Snellen lines or more than 0.3 logMAR, according to the method of recording visual acuity, or loss of light perception).
 - Reported by three studies (Babighian 2010; Falkenberry 2020; Yin 2024).
- Bleeding, as recorded by the investigators.
 - Reported by all studies.
- Endophthalmitis, as recorded by the investigators.
 - Reported by four studies (Babighian 2010; Kicinska 2023; Quaranta 1999; Sato 2018).
- Postoperative IOP spikes.
 - Reported by all studies except two (Maheshwari 2023; Quaranta 1999).

Excluded studies

After evaluating the full-text reports or trial registry records, we excluded 39 studies. Seventeen studies had an ineligible study design, 16 studies had an ineligible intervention, four studies had an ineligible unit of analysis, and two studies had an ineligible comparator, either comparing MIGS in both intervention groups or comparing different variations of the same MIGS to each other ([Supplementary material 3](#)).

Studies awaiting classification and ongoing studies

We assigned 11 trials as awaiting classification because they did not clearly mention if the intervention was MIGS, no published results are available, or the trials were terminated for various reasons by the study investigators. See [Supplementary material 4](#) for further information.

We have identified 15 registered ongoing studies ([Supplementary material 5](#)).

Risk of bias in included studies

Using the RoB 2 tool, we assessed risk of bias in the included studies for five outcomes in the summary of findings tables. These outcomes are the proportion of participants medication-free after surgery, the mean change in IOP, the mean change in the number of IOP-lowering drops taken per day, the proportion of participants who required further glaucoma surgery, and the proportion experiencing intra- or postoperative complications. Overall, we found all but one study (Babighian 2010) to be at moderate to high risk of bias for all reported outcomes.

Detailed risk of bias evaluations and the rationale behind our judgements can be found in [Supplementary material 6](#) and [Supplementary material 10](#). Full consensus decisions for all

signalling questions across each domain from all 10 studies are available on the Open Science Framework (OSF) [73].

Proportion of participants who were medication-free

We had concerns about the overall risk of bias in nearly all studies reporting this outcome. Babighian 2010 was the only study with low risk of bias in all domains. Five studies had a high overall risk of bias (Falkenberry 2020; Goldberg 2024; Sato 2018; Ting 2018; Yin 2024) and three studies had some concerns (Kicinska 2023; Maheshwari 2023; Quaranta 1999). The most common concerns were related to the randomisation process (insufficient detail on allocation concealment and occasional baseline imbalances) (Kicinska 2023; Maheshwari 2023; Quaranta 1999; Sato 2018; Yin 2024) and deviations from intended interventions, particularly where analyses were per-protocol (Goldberg 2024; Kicinska 2023; Yin 2024). We judged several studies at high risk of bias due to missing outcome data, including attrition and post-randomisation exclusions (Goldberg 2024; Sato 2018; Ting 2018; Yin 2024). In addition, some studies had increased risk of bias due to non-standardised or unclear criteria for stopping medications/defining surgical success (Falkenberry 2020; Goldberg 2024).

Rate of visual field progression/proportion of participants whose visual field loss progressed in the follow-up period

Two studies reported this outcome, but narratively (Babighian 2010; Kicinska 2023). Overall, this outcome was a high risk of bias in both studies, primarily because the studies did not report the outcome in an extractable form for our review (i.e. they did not provide numerical data on the proportion of participants with visual field progression per group, nor group-specific postoperative visual field MD values or changes from baseline). Instead, visual field findings were reported narratively. There were also some concerns due to deviations from intended interventions for Kicinska 2023 (analysis appeared to be per-protocol, rather than intention-to-treat), and due to the randomisation process because allocation concealment was not described.

Mean change in IOP

We had concerns about the overall risk of bias in all studies, for which this outcome was either reported or calculated by us, that were included in the summary of findings tables. Three studies had a high overall risk of bias (Goldberg 2024; Sato 2018; Ting 2018), while three studies had some overall concerns (Falkenberry 2020; Maheshwari 2023; Quaranta 1999). Three studies had concerns regarding the randomisation process due to insufficient reporting of randomisation or concealment methods and occasional baseline difference imbalances (Maheshwari 2023; Quaranta 1999; Sato 2018). We also had some concerns regarding bias introduced by deviations from intended interventions, in particular, where analyses were per-protocol or where protocol deviations were unclear (Maheshwari 2023; Quaranta 1999). We deemed the risk of bias for this outcome to be high for Goldberg 2024. We deemed three studies at high risk of bias due to missing outcome data, including attrition and post-randomisation exclusions (Goldberg 2024; Sato 2018; Ting 2018). Finally, five studies had concerns regarding bias in selection of the reported result because the mean change in IOP was not directly reported or because prespecified analysis plans were unavailable (Falkenberry 2020; Goldberg 2024; Maheshwari 2023; Quaranta 1999; Sato 2018).

Mean change in the number of IOP-lowering drops taken per day

We had concerns about the overall risk of bias in all studies, for which this outcome was either reported or calculated by us, that were included in the summary of findings tables. Four studies had a high overall risk of bias (Falkenberry 2020; Goldberg 2024; Ting 2018; Yin 2024), and one study had some concerns (Maheshwari 2023). Maheshwari 2023 had concerns regarding the randomisation process due to one group with a higher baseline number of medications. Yin 2024 had some concerns about randomisation due to no apparent methods of concealment. Maheshwari 2023 also had some concerns due to deviations from intended interventions, while Yin 2024 had a high risk of bias due to using a per-protocol analysis for missing data. We classified three studies as high risk of bias due to either missing outcome data or unclear handling of exclusions (Goldberg 2024; Ting 2018; Yin 2024). Three studies had some concerns regarding the measurement of the outcome (Falkenberry 2020; Goldberg 2024; Ting 2018), as postoperative medication adjustments and criteria for reducing or adding drops were not standardised or clearly defined. There were also some concerns about the selection of the reported result due to this lack of pre-specified protocols or criteria for the postoperative medication regimes (Falkenberry 2020; Goldberg 2024; Maheshwari 2023; Ting 2018).

Mean change in health-related quality of life (HRQoL), measured using any scale

This outcome was not reported by any study and therefore the RoB 2 tool could not be used.

Proportion of participants who required further glaucoma surgery including laser

We had concerns about the overall risk of bias in all studies reporting this outcome. We judged two studies as having some concerns overall (Kicinska 2023; Quaranta 1999), while four studies had an overall high risk of bias (Goldberg 2024; Sato 2018; Ting 2018; Yin 2024). Four studies had issues related to the randomisation process, due to insufficient reporting of the randomisation and/or allocation concealment methods (Kicinska 2023; Quaranta 1999; Sato 2018; Yin 2024). Two studies had some concerns regarding deviations from intended interventions where masking of participants and outcome assessors was unclear (Kicinska 2023; Quaranta 1999), and we judged this high risk in Goldberg 2024 and Yin 2024 due to a non-intention-to-treat analysis. Four studies had high risk of bias due to missing outcome data (Goldberg 2024; Sato 2018; Ting 2018; Yin 2024). Finally, three studies had some concerns about the selection of the reported result due to missing criteria for further surgery, lack of information in protocols, or because prespecified analysis plans were unavailable (Goldberg 2024; Quaranta 1999; Yin 2024).

Adverse events

All studies reported adverse events. Two studies had high overall risk of bias (Sato 2018; Ting 2018), while the remainder had some concerns overall. Sato had a high risk of bias regarding the randomisation process due to the aforementioned baseline differences between the groups and lack of mention of concealment methods. There were some concerns regarding the randomisation process in four studies due to not reporting on their method of concealment (Kicinska 2023; Maheshwari 2023;

Quaranta 1999; Yin 2024). There were some concerns regarding deviations from intended interventions in five studies (Goldberg 2024; Kicinska 2023; Maheshwari 2023; Quaranta 1999) where either masking of participants was unclear (Kicinska 2023; Quaranta 1999), participants did not receive the intended interventions (Yin 2024), or due to a non-intention-to-treat analysis (Goldberg 2024; Kicinska 2023; Maheshwari 2023). Two studies had a high risk of bias due to missing data (Sato 2018; Ting 2018) while eight studies had some concerns regarding the selection of reported results due to lack of protocols or no prespecified analysis plans being available (Arimura 2023; Babighian 2010; Falkenberry 2020; Goldberg 2024; Maheshwari 2023; Quaranta 1999; Sato 2018; Ting 2018).

Synthesis of results

Comparison 1: SC-MIGS (MIGS that act on the Schlemm's canal) versus non-MIGS surgical options

The results of this comparison are summarised in [Summary of findings 1](#). The certainty of evidence for all the outcomes in this comparison was either low or very low, as depicted in [Summary of findings 1](#). This was caused by a downgrade of two levels due to imprecision – there were a small number of events and wide confidence intervals for both the relative and absolute effect estimates. For some outcomes, this was further downgraded by another level due to the high risk of bias caused by some of the included studies.

There were three studies included in this comparison (Kicinska 2023; Quaranta 1999; Ting 2018). The interventions included were ab interno canaloplasty with iTrack combined with phaco (Kicinska 2023), ab interno goniotrabeculotomy (Quaranta 1999), and ab interno trabeculectomy with Trabectome combined with phaco (Ting 2018).

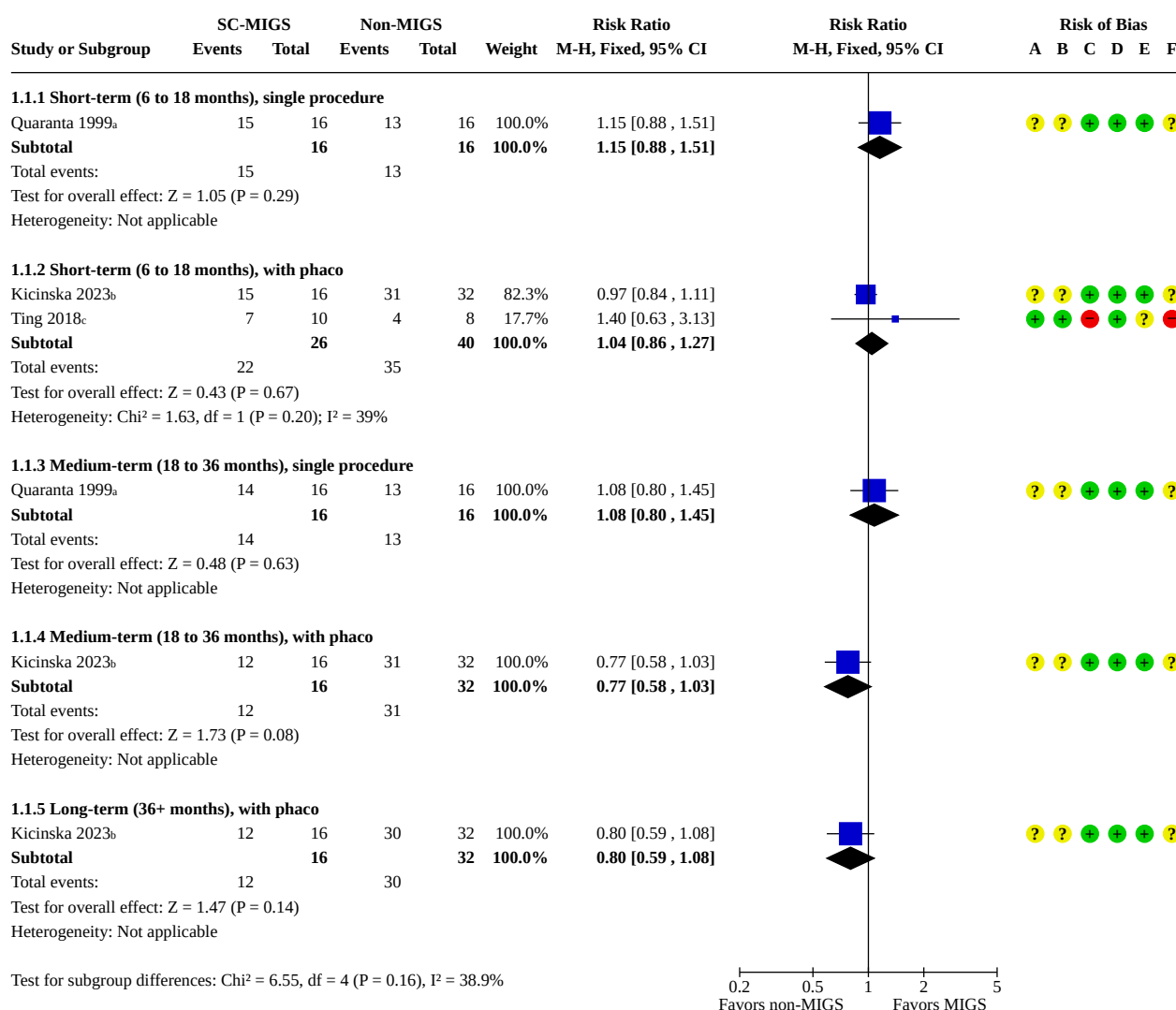
Critical outcomes

Proportion of participants medication-free

Short term (6 to 18 months)

Three studies reported the proportion of participants who were medication-free in the short term (6 to 18 months) (Kicinska 2023; Quaranta 1999; Ting 2018). These were divided into two subgroups depending on whether participants also received phacoemulsification as an intervention alongside the SC-MIGS.

Quaranta 1999 was the only study reporting on participants who were medication-free without phacoemulsification carried out alongside the SC-MIGS. There was no evidence of a difference in the group who had the SC-MIGS procedure compared with the non-MIGS group (RR 1.15, 95% CI 0.88 to 1.51; 1 study, 32 participants; low-certainty evidence; Analysis 1.1; [Figure 2](#)).

Figure 2. 1.1 Proportion of participants who were medication-free**Footnotes**^aAb interno Goniotrabeculotomy vs Trabeculectomy^bAb interno canaloplasty with iTrack + Phaco vs either Minicanaloplasty Ab Externo + Phaco OR Ab Externo Canaloplasty with iTrack + Phaco^cAb interno trabeculectomy with trabectome + Phaco vs Trabeculectomy + Phaco**Risk of bias legend**

(A) Bias arising from the randomization process

(B) Bias due to deviations from intended interventions

(C) Bias due to missing outcome data

(D) Bias in measurement of the outcome

(E) Bias in selection of the reported result

(F) Overall bias

Two studies reported the proportion of participants medication-free in the short term whose procedures were carried out alongside phacoemulsification (Kicinska 2023; Ting 2018). Meta-analysis of the data from these two studies resulted in no evidence of a difference between the SC-MIGS and the non-MIGS surgical procedures (RR 1.04, 95% CI 0.86 to 1.27; I² = 39%; 2 studies, 66 participants; very low-certainty evidence; Analysis 1.1; Figure 2).

Medium term (> 18 to 36 months)

Two studies reported the proportion of participants who were medication-free in the medium term. These were again divided into two subgroups depending on whether participants received phacoemulsification surgery as an intervention alongside the SC-MIGS procedure.

In one study, participants did not have phacoemulsification surgery alongside their SC-MIGS procedure (Quaranta 1999). Analysis showed no evidence of a difference in the proportion of participants who remained medication-free in the SC-MIGS and non-MIGS

groups (RR 1.08, 95% CI 0.80 to 1.45; 32 participants; low-certainty evidence; Analysis 1.1).

One study had participants have phacoemulsification alongside their SC-MIGS procedure (Kicinska 2023). Analysis showed no evidence of a difference in the proportion of participants who remained medication-free in the SC-MIGS group and the non-MIGS group (RR 0.77, 95% CI 0.58 to 1.03; 48 participants; low-certainty evidence; Analysis 1.1).

Long term (36+ months)

Only Kicinska 2023 reported outcomes for participants after more than 36 months. There were no reported differences between the SC-MIGS and non-MIGS groups (RR 0.80, 95% CI 0.59 to 1.08; 48 participants; low-certainty evidence; Analysis 1.1).

Rate of visual field progression/proportion of participants whose visual field loss progressed in the follow-up period

Only Kicinska 2023 reported overall visual field progression as an outcome. They reported that there was no evidence of a difference between the groups when comparing the MD and the Power Spectral Density (PSD) parameters. The certainty of the evidence was very low.

Important outcomes

Mean change in IOP

These studies were again divided in our analysis depending on whether the procedure was with phacoemulsification or without.

In only one study was the mean change in IOP at short-term follow-up (measured at 12 months) in participants with a single procedure calculated (Quaranta 1999). There was no evidence of a difference in the change in IOP between the SC-MIGS and the non-MIGS group (MD -0.75 mmHg, 95% CI -2.85 to 1.35; 1 study, 16 participants in each arm; low-certainty evidence; Analysis 1.2).

Single-study estimates from Ting 2018 showed no evidence of a difference in mean change of IOP at short-term follow-up (measured at 12 months) in participants who had a phacoemulsification surgery alongside their MIGS procedure (MD 3.70 mmHg, 95% CI -3.25 to 10.65; 1 study, 17 participants; very low-certainty evidence; Analysis 1.2).

Mean change in the number of IOP-lowering drops taken per day

Single-study estimates from Ting 2018 showed no evidence of a difference in the mean number of IOP-lowering drops taken per day (MD -0.71, 95% CI -1.79 to 0.37; 1 study, 16 participants; very low-certainty evidence; Analysis 1.3).

Proportion of participants with IOP of 14, 18, 21 mmHg or less in the short term

Because of the clinical heterogeneity, Analysis 1.4 includes the following separate subgroups:

Target IOP 14 mmHg or less, single procedure

There was only one study in this category. Quaranta 1999 reported the proportion of participants with an IOP of 14 mmHg or less. There was no evidence of a difference between the SC-MIGS and the non-MIGS group (RR 1.15, 95% CI 0.88 to 1.51; 1 study, 16 participants in each group; very low-certainty evidence).

Target IOP 14 mmHg or less, with phacoemulsification

Based on one study (Ting 2018), there was no evidence of a difference between the SC-MIGS and non-MIGS group in the proportion of participants with an IOP of 14 mmHg or less (RR 0.30, 95% CI 0.04 to 2.31; 1 study, 9 participants in the SC-MIGS arm, 8 participants in the non-MIGS arm; very low-certainty evidence).

Target IOP 18 mmHg or less, with phacoemulsification

Based on one study (Ting 2018), there was no evidence of a difference between the SC-MIGS and non-MIGS group in the proportion of participants with an IOP of 18mmHg or less (RR 1.07, 95% CI 0.53 to 2.17; 1 study, 9 participants in the SC-MIGS group versus 8 participants in the non-MIGS group; very low-certainty evidence).

Target IOP of 21 mmHg or less, with phacoemulsification

Based on one study (Ting 2018), there was no evidence of a difference between the SC-MIGS and non-MIGS group in the proportion of participants with an IOP of 21 mmHg or less (RR 1.14, 95% CI 0.82 to 1.58; 1 study, 9 participants in the SC-MIGS group versus 8 participants in the non-MIGS group; very low-certainty evidence).

Mean change in health-related quality of life (HRQoL)

None of the included trials reported this outcome.

Proportion of participants who needed further glaucoma surgery, including laser

In studies where a single procedure (without phacoemulsification) was carried out, only one study reported this outcome (Quaranta 1999). There were no participants in either group who had further glaucoma surgery (Analysis 1.5)

In studies where there was a glaucoma procedure with phacoemulsification in addition, two studies reported this outcome (Kicinska 2023; Ting 2018). There was no evidence of a difference in the proportion of participants who required further surgery in the SC-MIGS group compared with the non-MIGS group (RR 2.70, 95% CI 0.13 to 58.24; 2 studies, 25 participants in the SC-MIGS group versus 40 in the non-MIGS group; very low-certainty evidence; Analysis 1.5).

Adverse effects

Proportion of participants with loss of visual acuity

None of the included studies for this comparison included this outcome.

Proportion of participants with bleeding

One study reported the proportion of participants who experienced bleeding either during or following their surgery (Quaranta 1999). There was no evidence of a difference in the proportion of participants experiencing bleeding in the SC-MIGS group compared with the non-MIGS group (RR 13.0, 95% CI 0.79 to 213.09; 1 study, 16 participants in each group; low-certainty evidence; Analysis 1.6).

For studies where participants had phacoemulsification carried out alongside the glaucoma procedure (Kicinska 2023; Ting 2018), there was no evidence of a difference in the proportion of participants with bleeding between the SC-MIGS and the non-MIGS groups (RR

1.13, 95% CI 0.78 to 1.62; $I^2 = 49\%$; 2 studies, 25 participants in the SC-MIGS group compared with 40 in the non-MIGS group; very low-certainty evidence; Analysis 1.6). Bleeding was defined as hyphema in two studies (Quaranta 1999; Ting 2018) whereas in Kicinska 2023, this also included participants with micro-hyphema and vitreous haemorrhage.

Proportion of participants with IOP spikes

Two studies, in which included participants who had phacoemulsification carried out alongside the glaucoma procedure, reported IOP spikes (Kicinska 2023; Ting 2018). There was no evidence of a difference between the two groups in the proportion of participants with IOP spikes (RR 1.44, 95% CI 0.58 to 3.54; $I^2 = 0\%$; 2 studies, 25 participants in the SC-MIGS group versus 40 participants in the non-MIGS group; very low-certainty evidence; Analysis 1.7). The definitions of IOP spikes varied between the studies. Ting 2018 reported this as an increase in the IOP of 6 mmHg or more from baseline on day 1. In Kicinska 2023, the authors defined IOP spikes as an IOP of greater than 30 mmHg but did not clearly specify at what time point.

Proportion of participants with endophthalmitis

For studies with a single intervention, only Quaranta 1999 reported this outcome. There were no participants in either group with endophthalmitis.

For studies where participants also had phacoemulsification carried out alongside the glaucoma procedure, only Kicinska 2023 reported this outcome. There were no participants who experienced endophthalmitis.

Comparison 2: SC-MIGS + phacoemulsification (SC-MIGS) versus phacoemulsification only (phaco)

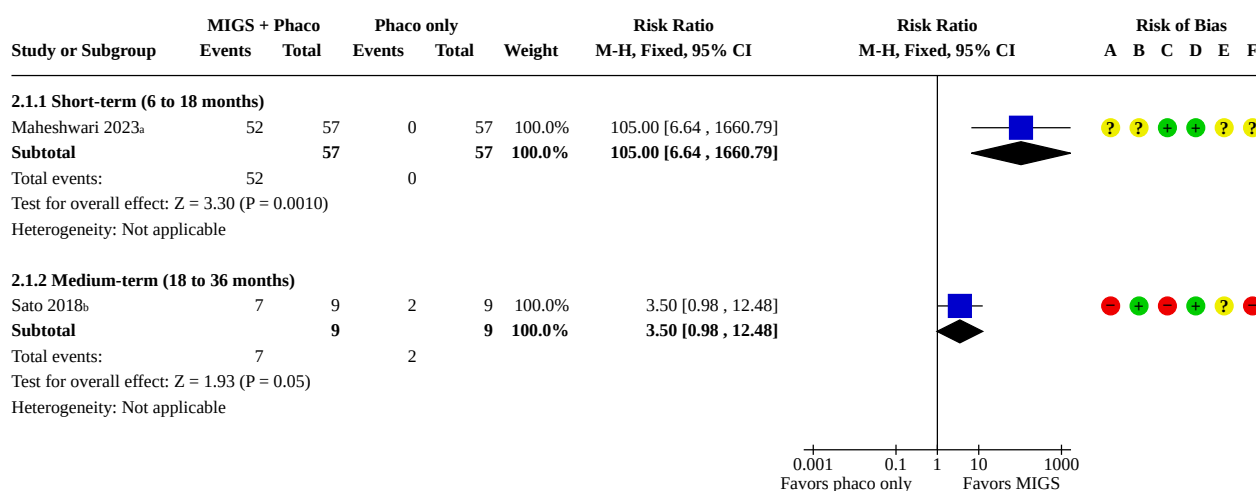
The results of this comparison are summarised in [Summary of findings 2](#). Further reasoning for the level of evidence is also presented in the table. There were two studies included in this comparison (Maheshwari 2023; Sato 2018). The interventions were ab interno Tanito microhook trabeculotomy + phaco (Maheshwari 2023) and 360 degree suture trabeculotomy ab interno + phaco (Sato 2018).

Critical outcomes

Proportion of participants medication-free

There are two studies that reported this outcome but at different post-procedure time points (Analysis 2.1; [Figure 3](#)).

Figure 3. 2.1 Proportion of participants who were medication-free



Footnotes

^aAb interno Tanito microhook trabeculotomy + Phaco with Phaco vs Phaco only, at 12 months

^b360-degree suture trabeculotomy ab interno + Phaco vs Phaco only, at 24 months

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Short term (6 to 18 months)

Only one study reported this outcome (Maheshwari 2023). There was a larger proportion of participants who were medication-free in the short term (12 months) in the SC-MIGS group compared with

the phaco group (RR 105.00, 95% CI 6.64 to 1660.79; 1 study, 114 participants; low-certainty evidence).

Medium term (> 18 to 36 months)

Minimally invasive trabecular meshwork surgery for open-angle glaucoma (Review)

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Only one study reported this outcome (Sato 2018). There was no evidence of a difference between the groups in the proportion of participants who were medication-free in the medium term (RR 3.50, 95% CI 0.98 to 12.48; 1 study, 9 participants in each group; very low-certainty evidence).

Rate of visual field progression/proportion of participants whose visual field loss progressed in the follow-up period

None of the included trials reported this outcome.

Important outcomes

Mean change in IOP

There were two studies that reported this outcome (Analysis 2.2).

Short term (6 to 18 months)

This outcome was calculated for one study in this category (Maheshwari 2023). Overall, there may be a significant mean difference between the groups in the mean change in IOP, favouring the SC-MIGS group at 12 months (MD -8.70 mmHg, 95% CI -10.12 to -7.28; 1 study, 57 participants in each group; low-certainty evidence).

Medium term (> 18 to 36 months)

One study reported this outcome at 24 months (Sato 2018). There was no evidence of a difference in the mean change in IOP between the groups (MD -3.80 mmHg, 95% CI -7.66 to 0.06; 1 study, 6 participants in the SC-MIGS group and 8 participants in the phaco-only group; low-certainty evidence).

Mean change in the number of IOP-lowering drops taken per day

One study reported this outcome in the short term (Maheshwari 2023). There was no evidence of a difference in the mean change in IOP-lowering drops per day (MD -0.10, 95% CI -0.39 to 0.19; 1 study, 57 participants in each group; low-certainty evidence; Analysis 2.3).

Proportion of participants with IOP of 18 mmHg or less in the short term

One study reported this outcome (Maheshwari 2023). There was a larger proportion of participants with a target IOP of 18 mmHg or less in the SC-MIGS group compared with the phaco-only group (RR 1.86, 95% CI 1.41 to 2.45; 1 study, 57 participants in each group; high-certainty evidence; Analysis 2.4).

Mean change in health-related quality of life (HRQoL)

None of the included trials reported this outcome.

Proportion of participants who needed further glaucoma surgery, including laser

Only one study reported the proportion of participants who required further glaucoma surgery (Sato 2018). There was no evidence of a difference in the proportion of participants who required glaucoma surgery in the phaco-only group compared with the SC-MIGS group (RR 0.33, 95% CI 0.02 to 7.24; 1 study, 9 participants in each group; very low-certainty evidence; Analysis 2.5).

Adverse effects

Proportion of participants with loss of visual acuity

Sato 2018 reported that no participants in either group had a decrease in their visual acuity by two or more lines.

Proportion of participants with bleeding

Both studies in this comparison reported this outcome (Maheshwari 2023; Sato 2018). When pooled, there was an increased risk of bleeding in participants who had SC-MIGS (RR 13.42, 95% CI 1.86 to 96.98; $I^2 = 0\%$; 2 studies, 66 participants in each group; low-certainty evidence; Analysis 2.7). Both studies defined bleeding as hyphema (Maheshwari 2023; Sato 2018). The certainty of evidence for this outcome is low.

Proportion of participants with IOP spikes

One study reported this outcome (Sato 2018). There was no evidence of a difference between the groups in the proportion of participants with IOP spikes (RR 1.00, 95% CI 0.27 to 3.69; 1 study, 9 participants in each group; very low-certainty evidence; Analysis 2.8). They defined an IOP spike as an IOP value of ≥ 30 mmHg within one month after the original surgery and a day-to-day increase in the IOP value of ≥ 5 mmHg.

Maheshwari 2023 reported that IOP spikes (defined as an IOP of ≥ 21 mmHg in the early postoperative period) were significantly more common in the phaco-only group, but did not report their figures.

Proportion of participants with endophthalmitis

One study reported this outcome (Sato 2018). It reported zero cases of endophthalmitis across both groups throughout the study period.

Comparison 3: SC-MIGS (Schlemm's canal MIGS) versus MIGS that do not act on the Schlemm's canal (non-SC-MIGS)

The results of this comparison are summarised in [Summary of findings 3](#). Two studies were included in this analysis, both comparing SC-MIGS with trabecular microbypass surgery with iStent and phacoemulsification. The interventions were ab interno canaloplasty with STREAMLINE system with phacoemulsification (Goldberg 2024) and Kahook Dual Blade goniotomy and phacoemulsification (Falkenberry 2020).

Critical outcomes

Proportion of participants medication-free in the short term (6 to 18 months)

There was no evidence of a difference in the proportion of participants medication-free in the short term (at 12 months) between the two groups (RR 1.07, 95% CI 0.92 to 1.24; $I^2 = 0\%$; 2 studies; 112 participants in the SC-MIGS group versus 115 participants in the non-SC-MIGS group; very low-certainty evidence; Analysis 3.1).

Rate of visual field progression/proportion of participants whose visual field loss progressed in the follow-up period

The included studies did not report this outcome.

Important outcomes

Mean change in IOP in the short term (6 to 18 months)

We manually calculated this outcome using the method outlined in the [Dealing with missing data](#) section of the methods. There was no evidence of a difference in the mean change in IOP in the SC-MIGS group compared with the non-SC-MIGS group (MD -0.25, 95% CI -1.13 to 0.63; $I^2 = 55\%$; 2 studies, 112 participants in the SC-MIGS group versus 115 participants in the non-SC-MIGS group; very low-certainty evidence; Analysis 3.2).

Mean change in the number of IOP-lowering drops taken per day in the short term (6 to 18 months)

We manually calculated this outcome using the method outlined in the [Dealing with missing data](#) section of the methods. There was no evidence of a difference in the mean change of IOP-lowering drops taken per day in the SC-MIGS group compared with the non-SC-MIGS group (MD -0.15, 95% CI -0.34 to 0.03; 2 studies, 112 participants in the SC-MIGS group versus 115 participants in the non-SC-MIGS group; very low-certainty evidence; Analysis 3.3).

Mean change in health-related quality of life (HRQoL)

None of the included trials reported this outcome.

Proportion of participants who needed further glaucoma surgery, including laser

Only Goldberg 2024 reported this outcome. There were no participants in either group who required further glaucoma surgery or laser during their follow-up period.

Adverse effects

Proportion of participants with loss of visual acuity

Falkenberry 2020 reported that there were no participants in either group who had worsening visual acuity of two lines or more.

Proportion of participants with bleeding

There may be evidence of an increased proportion of participants experiencing bleeding (defined as hyphema in both studies) in the SC-MIGS group compared with the non-SC-MIGS group (RR 3.10, 95% CI 0.50 to 19.36; 2 studies, 117 participants in the SC-MIGS group versus 121 participants in the non-SC-MIGS group; low-certainty evidence; Analysis 3.6).

Proportion of participants with IOP spikes

Falkenberry 2020 defined this as an IOP increase of 10 mmHg or greater from baseline at any point during the follow-up schedule. Goldberg 2024 defined this as an IOP increase of 10 mmHg or greater from baseline one month following the intervention.

There was no evidence of a difference in the proportion of participants with an IOP spike in the SC-MIGS group compared with the non-SC-MIGS group (RR 0.99, 95% CI 0.64 to 1.52; 2 studies, 117 participants in the SC-MIGS group versus 121 participants in the non-SC-MIGS group; moderate-certainty evidence; Analysis 3.7).

Proportion of participants with endophthalmitis

Only Falkenberry 2020 reported this outcome. There were no participants in either group who developed endophthalmitis.

Comparison 4: SC-MIGS1 versus SC-MIGS2

Two studies were eligible for this analysis, which was a comparison between two minimally invasive glaucoma surgery techniques, both of which acted on the Schlemm's canal (Arimura 2023; Yin 2024). Arimura 2023 had a different primary outcome (mean cross-sectional area of the incision sites of the trabecular meshwork) compared with the other studies. Therefore, the only other applicable outcomes to our review were the change in IOP (which they reported as a percentage change with interquartile ranges) and adverse event outcomes.

The interventions and comparisons in the included studies were also different to each other, which meant that pooling of the data or meta-analysis was inappropriate.

The included interventions and comparators for this comparison were Kahook Dual Blade goniotomy and phaco (KDB-Phaco) compared with ab interno microhook trabeculotomy and phaco (Microhook-Phaco) (Arimura 2023) and ab interno canaloplasty with iTrack (ABiC) compared with gonioscopy-assisted transluminal trabeculotomy (GATT) (Yin 2024).

The certainty of evidence for all of the below outcomes was very low, except for the proportion of participants experiencing the adverse effect of bleeding. For this outcome, the certainty of the evidence was low.

Critical outcomes

Proportion of participants medication-free in the short term (6 to 18 months)

Only Yin 2024 reported this outcome. There was no evidence of a difference between the two groups for the proportion of participants medication-free at 12 months follow-up (RR 0.73, 95% CI 0.52 to 1.03; 1 study, 71 participants, Analysis 4.1). Yin 2024 reported the reason for the reintroduction of IOP-lowering medication following surgery as (1) if the IOP exceeded 25 mmHg, or (2) if the IOP was between 21 mmHg and 25 mmHg on two consecutive measurements taken within three days of each other.

Rate of visual field progression/proportion of participants whose visual field loss progressed in the follow-up period

None of the studies in this domain reported this outcome.

Important outcomes

Mean change in IOP in the short term (6 to 18 months)

Arimura 2023 reported that there was no evidence of a difference between the two groups in the rate of change of IOP over 12 months.

Yin 2024 showed that there was no evidence of a difference in the mean change in IOP at 12 months between their two groups (MD 2.20, 95% CI -2.73 to 7.13; 1 study; 71 participants; Analysis 4.2).

Mean change in the number of IOP-lowering drops taken per day in the short term (6 to 18 months)

We manually calculated the mean change in the number of IOP-lowering drops taken per day for Yin 2024 as per the calculation outline in [Dealing with missing data](#). There was no evidence of a difference in mean change of IOP-lowering drops taken per day between the two groups (MD 0.40, 95% CI -0.11 to 0.91; 1 study, 71 participants; Analysis 4.3).

Proportion of participants who achieved an IOP of 15, 18, 21 mmHg or less in the short term (6 to 18 months)

Only Yin 2024 reported this outcome. There was no evidence of a difference in the proportion of participants who had an IOP of 15 mmHg or less (RR 0.51, 95% CI 0.25 to 1.05; 1 study, 71 participants), or the proportion of participants who had an IOP of 21 mmHg or less (RR 0.90, 95% CI 0.75 to 1.08; 1 study, 71 participants) (Analysis 4.4).

However, there was a greater proportion of participants who had an IOP of 18 mmHg or less following gonioscopy-assisted transluminal trabeculotomy (GATT) compared with ab interno canaloplasty (AbiC) (RR 0.60, 95% CI 0.41 to 0.88; 1 study, 71 participants; Analysis 4.4).

Mean change in health-related quality of life (HRQoL)

The included studies did not report this outcome.

Proportion of participants who needed further glaucoma surgery, including laser

Only Yin 2024 reported this outcome. There was no evidence of a difference in the proportion of participants who required further surgery after 12 months between the groups (RR 3.09, 95% CI 0.34 to 28.26; 1 study, 71 participants; Analysis 4.5). Arimura 2023 did not report this outcome.

Adverse effects

Proportion of participants with loss of visual acuity

Only Yin 2024 reported this outcome. There was no evidence of a difference in the proportion of participants who required further surgery after 12 months between the groups (RR 3.08, 95% CI 0.13 to 73.26; 1 study, 77 participants; Analysis 4.6)

Proportion of participants with bleeding

Both studies reported this outcome (Arimura 2023; Yin 2024).

A lower proportion of participants may have experienced bleeding in the ABiC group compared with the GATT group (RR 0.54, 95% CI 0.38 to 0.78; 1 study, 77 participants; Analysis 4.7).

There was no evidence of a difference in the KDB-Phaco group compared with the Microhook-phaco group (RR 1.67, 95% CI 0.44 to 6.34; 1 study, 58 participants; Analysis 4.7)

Bleeding was reported as a hyphema in both studies.

Proportion of participants with IOP spikes

Both studies reported this outcome. There was no difference between the groups in either study without phaco (RR 0.86, 95% CI 0.53 to 1.41; 1 study, 77 participants; Analysis 4.8) or with phaco (RR 0.67, 95% CI 0.21 to 2.12; 1 study, 58 participants; Analysis 4.8).

IOP spikes were defined by Yin 2024 as an IOP of greater than or equal to 30 mmHg within one month of surgery, whereas Arimura 2023 did not define this.

Proportion of participants with endophthalmitis at 12 months

The included studies did not report this outcome.

Comparison 5: SC-MIGS versus laser

There was only one study included in this analysis. Babighian 2010 compared the use of excimer laser trabeculotomy (ELT), which was

SC-MIGS, to 180 degree selective laser trabeculoplasty (SLT), which was laser.

Critical outcomes

Proportion of participants who were medication-free in the medium term (18 to 36 months)

There was no evidence of a difference in the proportion of participants medication-free in the medium term (at 24 months) between the two groups (RR 1.33, 95% CI 0.61 to 2.91; 1 study, 30 participants; Analysis 5.1). The certainty of the evidence for this outcome was moderate.

Rate of visual field progression/proportion of participants whose visual field loss progressed in the follow-up period

Babighian 2010 reported that there was no change in the visual field outcome at the end of the follow-up period, but no data were provided for this outcome.

Important outcomes

Mean change in IOP in the medium term (18 to 24 months)

We manually calculated this outcome, using the method outlined in the [Dealing with missing data](#) section of the methods. There may be a significantly greater mean change in IOP in the ELT group compared with the SLT group after 24 months (MD -2.60, 95% CI -3.91 to -1.29; 1 study, 30 participants; Analysis 5.2). The certainty of evidence for this outcome was low.

Mean change in the number of IOP-lowering drops taken per day in the medium term (18 to 24 months)

The included study did not report this outcome, nor report enough data for this to be calculated manually.

Mean change in health-related quality of life (HRQoL)

The included study did not report this outcome.

Proportion of participants who needed further glaucoma surgery, including laser

The included study did not report this outcome.

Adverse effects

Proportion of participants with loss of visual acuity

The included study did not report loss of visual acuity as an adverse effect.

Proportion of participants with bleeding

Only Babighian 2010 reported this outcome. There was a significantly higher proportion of participants experiencing bleeding in the ELT group compared with the SLT group. No participants from the SLT group experienced any bleeding (RR 25.0, 95% CI 1.61 to 387.35; 1 study, 30 participants; Analysis 5.3). The certainty of evidence for this outcome was low.

Proportion of participants with IOP spikes

Only Babighian 2010 reported this outcome. There was no evidence of a difference in the proportion of participants experiencing IOP spikes between the two groups (RR 1.50, 95% CI 0.29 to 7.73; 1 study, 30 participants; Analysis 5.4). An IOP spike was defined as an IOP rise of > 5 mmHg postoperatively. All participants had their IOP

spikes resolved within 48 hours. The certainty of evidence for this outcome was moderate.

Proportion of participants with endophthalmitis

Babighian 2010 reported that no participants in either group had endophthalmitis.

DISCUSSION

Summary of main results

We included 10 completed parallel-group RCTs involving a total of 639 eyes from 639 participants. By the end of the follow-up period across all included studies, outcome data were available for 605 eyes corresponding to 605 participants. We assessed study outcomes as generally having mostly low- to very low-certainty evidence with some outcomes having a moderate certainty of evidence.

Analysis of the critical outcomes showed that, in the short term (6 to 18 months), there may be a greater proportion of participants who were medication-free in those who underwent minimally invasive glaucoma surgery opening the trabecular meshwork or enlarging Schlemm's canal (SC-MIGS) combined with phacoemulsification compared with those who received phacoemulsification alone, though the certainty of evidence for this outcome was low. However, this difference was not observed in the medium term (18 to 36 months). There was no evidence of a difference in the proportion of participants who were medication-free when comparing SC-MIGS with non-minimally invasive glaucoma surgery, other SC-MIGS procedures, non-SC-MIGS procedures, or laser, though again the certainty of evidence for these outcomes was low, apart from the comparison with laser, which had a moderate certainty of evidence.

For the important outcomes, there may be a difference in the mean change in intraocular pressure (IOP) for SC-MIGS compared with phacoemulsification alone, and compared with laser. The certainty of evidence for this outcome ranged from low to very low. However, there was no evidence of a difference when comparing SC-MIGS with other SC-MIGS procedures or non-minimally invasive glaucoma surgery. The evidence also suggested little to no difference in the mean change in the number of IOP-lowering drops taken per day between SC-MIGS and other comparators (non-minimally invasive glaucoma surgery, phacoemulsification alone, other SC-MIGS, or non-SC-MIGS). The certainty of evidence for this outcome ranged from moderate to very low. Similarly, the evidence suggested little to no difference in the proportion of participants requiring further surgery or laser, although the certainty of the evidence for this outcome was very low.

With regard to adverse events, only three studies reported loss of visual acuity as an outcome (Babighian 2010; Falkenberry 2020; Yin 2024), and there was no evidence of a difference between the intervention and the comparator groups for this. For rates of endophthalmitis, there was no evidence of a difference between SC-MIGS and comparators including non-minimally invasive glaucoma surgery, phacoemulsification alone, or laser. SC-MIGS may be associated with an increased risk of bleeding compared with phacoemulsification alone (low-certainty evidence) and laser (low-certainty evidence); however, when compared with non-SC-MIGS or non-minimally invasive glaucoma surgery, the evidence suggested little to no difference in bleeding risk, though

the certainty was low or very low. For postoperative IOP spikes, there was little to no difference between SC-MIGS and non-SC-MIGS or laser, and no clear evidence of a difference compared with non-minimally invasive glaucoma surgery or phacoemulsification alone, although the evidence was again very uncertain for the comparison with non-SC-MIGS, and moderate for the comparison with laser.

Limitations of the evidence included in the review

Several of the studies in this review did not utilise the standardised method of reporting outcomes as recommended by the World Glaucoma Association (WGA) [45]. The WGA recommends that IOP success should be reported with appropriate upper limits, e.g. the proportion of participants with an IOP of ≤ 21 , 18, 15, or 12 mmHg. They also recommended definitions of complete success, which would correspond to participants with sufficient IOP control without any additional IOP-lowering medication. Visual field progression, either in the form of the proportion of participants who had progression of their visual fields, or the mean deviations pre- and postoperatively, is also recommended to be reported. Despite these guidelines, several studies did not report these outcomes, which limited our ability to perform appropriate meta-analyses of the data.

We categorised the certainty of evidence in this review as very low, low, moderate, or high. The main reasons for downgrading the evidence were primarily due to imprecision, caused by the small number of events and wide confidence intervals for both the relative and absolute effect estimates. Additionally, there was significant risk of bias in some studies, which further reduced the certainty in several of the analyses. Several concerns were identified regarding the risk of bias in the studies included, mainly regarding the size of the sample, usage of per-protocol analyses, and high relative rates of loss to follow-up. We did not use funnel plots in this review because each outcome measured had fewer than 10 studies included in the analysis.

Out of the included completed studies, three had no declared sources of funding (Babighian 2010; Maheshwari 2023; Sato 2018), four studies received funding from either government or university grants (Arimura 2023; Kicinska 2023; Quaranta 1999; Ting 2018), and two studies were funded by industry (Falkenberry 2020; Goldberg 2024). There are 15 ongoing studies identified by the review.

This review divided studies into five comparisons that reflect distinct areas of clinical equipoise (i.e. augmentation to phacoemulsification, alternatives to trabeculectomy, alternatives to laser, and inter-MIGS comparisons) and correspond to different patient populations, baseline risks, and treatment intents. As a result, each comparison comprised three or fewer studies, limiting the strength of conclusions and comparisons between the different interventions from the available data. Consequently, the evidence compiled in this review may prove difficult to translate meaningfully into clinical practice. The anticipated publication of additional studies on the efficacy of MIGS should provide further data, allowing for more robust conclusions to be drawn once more data can be pooled together for meta-analysis.

In terms of the external validity of the studies involved in this review, while a range of populations are covered, including from Europe, America, China, and India, there remains a limitation on the results from participants from an African nation or population.

This group is often under-represented in glaucoma clinical trials despite African descent populations having the highest prevalence of glaucoma amongst ethnic groups [74, 75, 76]. Therefore, there are concerns about whether results from the included studies can be generalised to this population.

Furthermore, there is limited evidence on the cost-effectiveness of MIGS. It is unclear if the cost of performing MIGS is outweighed by cost savings through decreased medication and less future intervention. In developed countries, where the cost can be compared with more expensive preservative-free glaucoma medication, MIGS may prove to be cost-effective. However, in developing countries, these procedures can be expensive for various reasons, including the cost of devices, poor coverage by insurance, and a lack of trained surgeons. Furthermore, even if MIGS were to take a foothold in developing countries, providers, patients, and insurers may choose the cheapest efficacious MIGS, rather than choosing one that may be more expensive, even if it provides better results, especially if those results are marginal. On the other hand, it is important to consider that traditional glaucoma surgery and medical treatment with eye drops for life may also be difficult in developing countries due to similar concerns [77, 78, 79].

In terms of safety profile, compared with some other comparator groups, SC-MIGS procedures may be associated with an increased risk of bleeding. This was almost always in the form of postoperative hyphema, which (as several studies have noted) is often non-significant. In terms of other adverse events, there were no notable differences between SC-MIGS and comparators, whether with or without phacoemulsification. However, confidence intervals were wide, and the reporting of adverse events between studies was also not homogeneous. For example, the definition of IOP spikes varied greatly between studies.

Further research in this field will enable more robust conclusions as to not only the efficacy and cost-effectiveness of SC-MIGS compared with other treatments, but also the safety profile.

Limitations of the review processes

The information specialist performed a highly sensitive search to identify potentially eligible studies. This included the standard Cochrane review methodology to minimise bias, and we followed the MECIR (Methodological Expectations of Cochrane Intervention Reviews) standards for conducting this review. None of the authors in this review have any financial conflicts of interest related to the review topic, nor direct involvement in the conduct, analysis, or publication of the clinical trials related to this review.

All studies included in this review reported outcomes in the follow-up window specified by this review's protocol (more than six months following intervention). We deemed another study to be comparing two variants of the same MIGS procedure to each other and therefore to be ineligible for inclusion (Sato 2021 [80, 81]).

Several studies did not include outcome data for 'Mean change in IOP' and 'Mean change in number of IOP-lowering medications', two important outcomes that we previously specified in our study protocol. Therefore, we manually calculated these ourselves as outlined in the methods of this review. For the calculation of the standard deviation, due to the limited availability of existing evidence in the literature for the imputed correlation coefficient for

each intervention, the value of 0.5 was chosen as suggested by a senior author (GV).

Agreements and disagreements with other studies or reviews

MIGS versus non-MIGS

A review examining IOP and glaucoma medications following ab interno trabecular bypass surgery with Trabectome (AIT) was published previously [82]. This review included only prospective or retrospective case series or cohorts with no randomised controlled trials included. The study found that while AIT does reduce intraocular pressure by 36%, reduces dependency on medications, and has excellent safety outcomes, a greater effect is seen with non-MIGS alternatives such as trabeculectomy. Our review includes one study looking at the effects of AIT (Ting 2018), but this is a very underpowered study, and when combined with other studies in Comparison 1 (MIGS versus non-MIGS), there was no evidence of a difference in the critical or important outcomes between the groups.

A systematic review by Zaifar et al. (2024) compared MIGS to trabeculectomy (non-MIGS) [83]. The authors found a comparable outcome between the two groups in terms of reduction in intraocular pressure; however, the study did suffer from low sample sizes. They noted that there were higher complication rates in the trabeculectomy group. However, in our analysis we found no evidence of a difference in the complication rates between the two groups.

MIGS with phaco versus phaco only

A review by Radwan et al. (2024) exploring the effects of the Kahook Dual Blade with phacoemulsification compared with phacoemulsification only showed that there was a significant decrease in IOP in patients who had the combined procedure compared with phaco only (22.74% compared with 9.62%), but with significant heterogeneity [84]. They also reported a significant reduction in the number of pressure-reducing eye drops in both groups and more so in the MIGS group (1.35 less compared with 0.36 less) but again with significant heterogeneity.

Another review by Yuan et al. (2024) looking at MIGS + phacoemulsification versus phacoemulsification only also showed a significant reduction in IOP in the MIGS + phaco group compared with the phaco-only group, but again their analyses showed significant heterogeneity in the results [85]. They also reported that common complications in the MIGS group were IOP spikes and the tendency to develop hyphema. As we described, they also report that several studies are lacking in consistent reporting of results as recommended by the World Glaucoma Association guidelines for surgical trials. The findings of Yuan et al. are in agreement with our conclusions from our Comparison 2 (MIGS + phaco versus phaco only) where there was a significant difference in IOP at 12 months in the MIGS + phaco group.

Neither of the above reviews included analyses on our critical outcomes of the proportion of participants medication-free or visual field progression.

Finally, a systematic review and meta-analysis by Oo et al. (2024) showed a significant reduction in IOP for participants who had Kahook Dual Blade (KDB) phaco [86]. The authors also looked at

iStents; however, they were unable to make any direct comparison between the two groups. The included Trabectome participants were deemed to be too small in number to perform any appropriate analysis.

There were no further systematic reviews that we found comparing a MIGS acting on the Schlemm's canal to another MIGS acting on the Schlemm's canal, nor MIGS acting on the Schlemm's canal compared with any other form of MIGS, nor any systematic review on MIGS versus laser.

AUTHORS' CONCLUSIONS

Implications for practice

Minimally invasive glaucoma surgeries (MIGS) have seen a rapid expansion globally as a treatment option for people with glaucoma. There are a wide range of MIGS available, and several factors, including the availability of technology and surgeon preference, play a role in the choice of intervention. However, it remains unclear which specific MIGS are more appropriate to be used, and when, in terms of safety and efficacy.

Based on our review, the following key clinical conclusions can be drawn from the current body of evidence regarding MIGS techniques that open the trabecular meshwork or enlarge Schlemm's canal (SC-MIGS):

- There may be little to no difference in the proportion of participants who are medication-free in the short, medium, or long term between SC-MIGS and non-MIGS surgical treatment, with or without phacoemulsification. The certainty of the underlying evidence is either low or very low.
- Compared with phacoemulsification alone, while SC-MIGS may statistically demonstrate a higher proportion of medication-free participants in the short term and a significantly greater short-term reduction in mean intraocular pressure (IOP), the certainty of the underlying evidence for all these conclusions is low, based on one study with a very wide confidence interval, indicating high imprecision, and therefore uncertainty of this finding.
- SC-MIGS may also be associated with a greater reduction in mean IOP compared with laser treatment. The certainty of evidence for this outcome is low.

There are concerns regarding the quality and heterogeneity of the included studies, particularly concerning interventions, comparators, and outcome reporting. Future high-quality research is necessary to validate these findings and establish clearer guidance for clinical practice.

Implications for research

The use of MIGS has increasingly become a standard therapeutic option in the management of glaucoma. Despite their growing popularity, the selection of specific MIGS procedures often depends on local surgical practices and clinician preference, influenced by perceived efficacy and familiarity. To date, there are no universally accepted guidelines that clearly define when and how MIGS should be used, nor which specific interventions should be considered first-line. This is further complicated by the limited number of randomised controlled trials (RCTs) directly comparing various MIGS procedures.

A key limitation in the current literature is the lack of standardised reporting of MIGS trials. Several studies in this review did not adhere to the standardised outcome reporting guidelines as established by the World Glaucoma Association (WGA) [45]. This inconsistency in methodology limits meaningful comparisons and the ability to perform a robust meta-analysis. Therefore, future research should adopt these standardised protocols for both study design and reporting.

Furthermore, several of the studies in this review were limited by the relatively short follow-up periods. Of the included studies, only Babighian 2010, Kicinska 2023, and Quaranta 1999 had a follow-up period extending beyond 12 months. However, considering that MIGS are a relatively new class of interventions, there will likely be longer-term outcome data in the future. In addition, we downgraded several of the outcomes in terms of certainty of evidence, often due to small sample sizes, risk of bias, or as a result of wide confidence intervals around both the relative and absolute estimates of effect.

Additionally, there is a need to incorporate a wider diversity of populations in MIGS studies. While there is a wide range of studies from several different countries, individual study populations were largely homogeneous. The lack of diversity is a significant limitation to the generalisability of the conclusions drawn from these studies, particularly when different studies had varying comparators and interventions. Furthermore, there is a pressing need for future research to be particularly focused on groups who are disproportionately affected by glaucoma, such as those of African descent, who also remain under-represented in research trials [76].

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015186.pub2](https://doi.org/10.1002/14651858.CD015186.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Data on study characteristics

Supplementary material 10 Risk of bias for rate of visual field progression (dB/time) - comparison 1/5

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

Cochrane Eyes and Vision supported the authors in the development of this review.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Tianjing Li, MD, MHS, PhD, University of Colorado Anschutz Medical Campus;
- Managing Editor (provided editorial guidance to authors, edited the article): Anupa Shah, Cochrane Editorial Service;
- Editorial Assistant (selected peer reviewers, conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Andrew Savage, Cochrane Editorial Service;
- Copy Editor (copy editing and production): Jenny Bellorini, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Clemens A. Strohmaier, MD, PhD (clinical and content review); Sheila Page (patient and public review); Nuala Livingstone and Katharina da Silva Lopes, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

Contributions of authors

KV and RG screened the search results and extracted the data for the review.

KV, RG, TWY, AB, and KH drafted the manuscript.

GV and CB aided with statistical analysis.

AB, KH, and GG contributed to critical revision of the manuscript.

Declarations of interest

KV: works as an ophthalmology resident doctor in the Thames Valley (Oxford) Deanery, UK.

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TWY: reports a grant from the National Eye Institute, National Institutes of Health, USA (payment to the institution); Cochrane Eyes and Vision Methodologist but was not involved in the editorial process for this review.

GV: is Co-ordinating Editor for Cochrane Eyes and Vision. All editorial decisions and sign off were made by another Editor.

AKB: works as a glaucoma surgeon at the University of Michigan Kellogg Eye Center and performs procedures described in the review.

CB: none known.

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GG: reports the following financial relationships: Consultant/Advisor, Lecture Fees/Speaker, and Research Support from Alcon Laboratories; Lecture Fees/Speaker from Bausch & Lomb; Consultant/Advisor and Research Support from Sight Sciences; Consultant/Advisor and Lecture Fees/Speaker from Vialase; Research Support from Oertli; and notes that their private practice encompasses this area. Investigating or has investigated: Hydrus; ELIOS laser; PMS; iStent; Omni.

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Registration and protocol

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Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library: full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies included ([Supplementary material 2](#)), ongoing ([Supplementary material 5](#)) or awaiting classification ([Supplementary material 4](#)), or excluded at the full-text screen ([Supplementary material 3](#)), in the final

review; study data, including study information, study arms, and study results ([Supplementary material 8](#)); consensus risk of bias assessments ([Supplementary material 6](#)); and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 7](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Covidence are available from the authors on reasonable request.

History

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ADDITIONAL TABLES

Table 1. Summary of included studies

Study ID	Study design (follow-up)	Setting	Intervention	Control	Number of eyes randomised	Population (mean age (SD); % male)	Outcomes with available data	Calculated outcomes
Arimura 2023	Parallel group (12 months)	1 hospital, Japan	Kahook Dual Blade goniotomy + phacoemulsification (KDB-Phaco)	Ab interno microhook trabeculotomy + phacoemulsification	60	KDB-Phaco: 72.9 ± 8.2 Micro-hook-Phaco: 72.3 ± 8.3; 40%	- Percentage change in IOP - Proportion of participants experiencing the complication of bleeding - Proportion of participants experiencing the complication of IOP spikes	
Babighian 2009	Parallel group (24 months)	1 hospital, Italy	Excimer laser trabeculotomy (ELT)	Selective laser trabeculoplasty 180 degrees (SLT)	30	65.5 (4.2); 43.3%	- Proportion of participants medication-free in the medium term - Proportion of participants experiencing the complication of bleeding - Proportion of participants experiencing the complication of IOP spikes - Proportion of participants experiencing the complication of endophthalmitis	- Mean change in IOP in the medium term - Mean change in number of IOP-lowering drops taken per day in the medium term
Yin 2024	Parallel group (12 months)	1 hospital, China	Canaloplasty ab interno	Gonioscopy assisted transluminal trabeculotomy (GATT)	80	40.8 (13.9); 74.0%	- Proportion of participants who were medication-free in the short term - Mean change in IOP in the short term - Proportion of participants who achieved an IOP of 15, 18, 21 mmHg or less - Proportion of participants who required further glaucoma surgery (including laser) - Proportion of participants experiencing the complication of loss of visual acuity - Proportion of participants experiencing the complication of bleeding	Mean change in the number of IOP-lowering drops taken per day in the short term

Table 1. Summary of included studies (Continued)

Falkenberry 2020	Parallel group (12 months)	8 centres, USA	Kahook Dual Blade goniotomy + phacoemulsification (KDB-Phaco)	Trabecular micro by-pass stent with iStent + phacoemulsification (iStent-Phaco)	166	KDB-Phaco 70.2 (SE 0.91) iStent-Phaco 71.1 (SE 0.88); 46.4%	• Proportion of participants who were medication-free in the short term • Proportion of participants experiencing the complication of bleeding • Proportion of participants experiencing the complication of IOP spikes	• Mean change in IOP in the short term • Mean change in the number of IOP-lowering drops taken per day in the short term
Kicinska 2022	Parallel group (36 months)	1 hospital, Poland	Canaloplasty ab interno + phacoemulsification (ABiC)	Either mini-canaloplasty ab externo + phacoemulsification (miniABeC) OR ab externo canaloplasty with iTrack + phacoemulsification (ABeC)	48	ABiC 74 (7) ABeC 77 (7) miniABeC 74 (8) ABiC 12.5% ABeC 25% miniABeC 37.5%	• Proportion of participants who were medication-free in the short and medium term • Proportion of participants who required further glaucoma surgery (including laser) • Proportion of participants experiencing the complication of bleeding • Proportion of participants experiencing the complication of IOP spikes • Proportion of participants experiencing the complication of endophthalmitis	
Quaranta 1999	Parallel group (24 months)	1 centre, Italy	Ab interno goniotrabeculotomy (AIGT)	Trabeculectomy with mitomycin C	32	AIGT 55.31 (6.8) Trabeculectomy 55.06 (8.5); not reported	• Proportion of participants who were medication-free in the short term • Proportion of participants who achieved an IOP of 14 mmHg or less • Proportion of participants experiencing the complication of bleeding • Proportion of participants experiencing the complication of endophthalmitis	• Mean change in IOP in the short term
Sato 2018	Parallel group (24 months)	1 centre, Japan	360 degree suture trabeculotomy	Phacoemulsification	18	Combined 74.2 (9.5)	• Proportion of participants medication-free in the medium term	• Mean change in IOP in the

Table 1. Summary of included studies (Continued)

			my ab interno (combined) + phacoemulsification (Phaco only)			Phacoemulsification only 74.4 (3.6); 44.4%	- Proportion of participants who required further glaucoma surgery (including laser) - Proportion of participants experiencing the complication of bleeding - Proportion of participants experiencing the complication of IOP spikes - Proportion of participants experiencing the complication of endophthalmitis	medium term
Ting 2018	Parallel group (12 months)	1 hospital, Canada	Ab interno trabeculotomy with Trabectome + phacoemulsification (Phaco-AIT)	Trabeculotomy + phacoemulsification (Phaco-Trab)	19	Phaco-AIT 71.3 (6.3) Phaco-Trab 67.4 (5.9); 52.6%	- Proportion of participants medication-free in the short term - Proportion of participants who achieved an IOP of 14, 18, 21 mmHg or less - Proportion of participants who required further glaucoma surgery (including laser) - Proportion of participants experiencing the complication of bleeding - Proportion of participants experiencing the complication of IOP spikes	- Mean change in IOP in the short term - Mean change in number of IOP-lowering drops taken per day in the short term
Maheswari 2023	Parallel Group (12 months)	1 hospital, India	Ab interno Tanito microhook trabeculotomy + phacoemulsification (microLOT)	Phacoemulsification (Phaco only)	114	MicroLOT 62.2 (8.4) Phaco only 63.1 (8.2); 62.3%	- Proportion of participants medication-free in the short term - Proportion of participants who achieved an IOP of 18 mmHg or less - Proportion of participants experiencing the complication of bleeding	- Mean change in IOP in the short term - Mean change in number of IOP-lowering drops taken per day in the short term
Goldberg 2024	Parallel group (6 months)	Multiple centres in the USA and Latin America	STREAMLINE canaloplasty + phacoemulsification	Trabecular micro bypass stent with iStent + phacoemulsification	72	STREAMLINE: 70.10 (7.67) iStent-Phaco 70.60 (7.07); 38.9	- Proportion of participants medication-free in the short term - Proportion of participants who required further glaucoma surgery (including laser)	- Mean change in IOP in the short term - Mean change in

	(iStent-Phaco)	%	• Proportion of participants experiencing the complication of bleeding • Proportion of participants experiencing the complication of IOP spikes	number of IOP-lowering drops taken per day in the short term
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IOP: intraocular pressure; mmHg: millimetres of mercury; Phaco: phacoemulsification; SD: standard deviation; SE: standard error